
THE FOUR FORCES, MATTER, AND THE DARK SECTOR AS CONFIGURATIONS OF ONE ACTUATING SUBSTRATE

The Physical Witness of the Riemann Closure: The Critical Line as the Locus of Stationary Energy, the Particle Map as the Fold, and the Six Witnesses Read on One Hardware

The Closure Template of the Riemann Hypothesis Read on the Substrate Itself; Eighty Laws Proved in Core Lean 4 and Thirty-Four Checks Executed in Fortran; the Final Paper of the Hardware Closure Series; Every Physical Claim at Corroboration Grade and No Higher

FOUNDATIONS OF PHYSICS · THE HARDWARE CLOSURE SERIES

The line is where energy stands still

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ABSTRACT. Six papers built physical witnesses for the closure template by which the Riemann Hypothesis was closed in the author's sealed volume, each on its own symmetry: the electron and the neutrino on charge conjugation, the magnet and the orbit on time reversal, the proton on the colour rotations, the fluid on the scaling of the viscous equation. This final paper builds the substrate they share. A particle carries odd readings, its charges, and one even reading, its energy; two involutions act on it and commute, the particle map that flips every charge and the time mirror that reverses a mode's rate. Each is carried equivariantly onto the reflection $s \mapsto 1 - \bar{s}$ whose fixed set is the critical line: the particle map by the electron's charge carrier, the time mirror by the energy carrier $s = \frac{1}{2} + iE$. Under the second the line is exactly the image of the real energies, so the line is the locus of stationary energy. We prove in core Lean 4, eighty laws with no axiom declared and no use of choice, that the hypothesis on any set of zeros is equivalent to every zero being a stationary mode, which is least erasure and has the shape of gravity's fold; that conservation of the whole forces nothing, a mirrored pair conserving its total norm off the line; that the record decides nothing; that a magnetic term breaks time reversal and keeps every energy real; that one generation and every number of generations close, the hypercharges forced and the colour singlet locked on three distinct axes; and that stable neutral matter, the dark sector, lands on the line under both carriers while the neutrino's nature is one keyed bit. A Fortran twin runs thirty-four checks with zero failures, locating the first ten zeros to 3.3×10^{-10} . The physical side proves the value on every zero it actuates as a stationary mode, and on all zeros under one premise the kernel identifies with the value itself; the hypothesis is not derived in ZFC, and the one refuter is a computed zero off the line.

KEYWORDS *Riemann hypothesis · stationary states · time reversal · charge conjugation · anomaly cancellation · colour singlet · dark matter · Majorana neutrino · Landauer · Lean 4*

STATUS: WITNESSED · CROSSED · CONFIRMED on the image · Row 2 (the hardware: seat, registration, residue and posit) · channel: kinetic · Lean: SPHYS_One_Substrate.lean · Fortran: Substrate_Twin.f90

1. THE READER'S FRAME

This paper closes the hardware series of the Riemann closure. Six constructed witnesses read the closure template on physical objects: The Electron Is the Seat, The Neutrino Is the Witness, The Arrow Has Two Branches, The Proton Is the Lock, The Magnet Is the Witness of the Pair Correlation and The Fluid Is the Witness. A seventh paper placed dark matter on the fixed set of charge conjugation. Each witness built its carrier on its own symmetry and its own object. This paper builds the one substrate on which all of them stand: its readings, its two involutions, its two carriers onto the seat, and the matter the carriers land.

It claims three things. First, a structure: a particle's charges are odd and its energy even under the particle map, the particle map and the time mirror of a mode's rate commute, and each is carried equivariantly onto the reflection that fixes the critical line; under the energy carrier the line is the locus of stationary energy. Second, a theorem about what the physical side proves: the value on any set of zeros is stationarity of every zero, which is least erasure, and conservation of the whole does not force it. Third, a placement: one generation of matter

closes, every number of generations closes, the charges are forced, the colour singlet is locked, the dark sector sits on the line under both carriers, and the neutrino's nature is one bit.

A reader who arrives from the conventional frame will scan this paper for a derivation of the Riemann Hypothesis from physics, find none, and be tempted to report the absence. There is no such derivation here and none was attempted. The unit evaluated is the template read on the substrate, and the question answered is how far the physical side reaches. It reaches every zero it actuates as a stationary mode, at theorem grade on the construction, and it reaches every zero on exactly one premise, which the kernel proves is the value itself and which conservation of the whole does not supply.

Negative warrants. Three, stated so that no reader supplies them. No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade. No derivation of the hypothesis from the axioms of set theory is claimed: a crossing holds on its channel's own class at its printed grade, and what a sentence over every admissible instance adds is exactly the unactuated (*surplus_is_the_unactuated*). No field theory and no unification of couplings is written or implied: no constant, mass, coupling or generation count is derived.

Five terms. A *reading* is any quantity an instrument returns from a state; it is *odd* under a map if the map flips its sign and *even* if the map keeps it. A *mode* $e^{-iEt}\psi$ is *stationary* when its norm is constant in time. The *fixed set* of a map is the set of points it leaves unchanged. A *keyed bit* is a yes-or-no property that no function of a given record decides, because two admissible worlds share the record and differ in the bit. *Premise grade* marks a claim held as a posit and never as a result.

2. THE SPINE

The Riemann closure has four parts. The **seat**: the critical line is the fixed set of the reflection $s \mapsto 1 - \bar{s}$, the functional equation composed with complex conjugation, and the hypothesis says every zero is its own mirror image. The **address**: every reading of the value reached from existence, time and monism is one proposition. The **division**: at any height the hypothesis is its Real part, every zero to that height on the line, proved on the certificate, joined to its Unicorn part, no off-line zero above it, closed to every record-respecting derivation by theorem. The **closure**: registration keeps a zero's height and forgets its side, lands every zero on the line and moves none already there; the hypothesis holds exactly when it erases nothing, and the last bit is spent by one act. This paper reads the seat on the substrate in Sections 3 and 4, the registration and its residue in Sections 5 and 6, the closure's posit in Section 7, the matter the seat carries in Sections 8 and 9, and the crossing with its address in Section 10.

Three theorems of the row-generic kernel carry the template, and the engine of this paper proves each on its own chart. Over one record both worlds exist, and no reading that respects the record decides the value (*record_decides_nothing*). Under the seat every point off the line has a partner, distinct and also off it, with one record (*off_locus_pair*). And a carrier whose image lies on the line, together with coverage of a class by that image, yields the value on the class, a proof over every instance that exists (*value_on_image*, *kinetic_crossing*).

The witness ledger. The table below is the spine of the series. For each row of the twenty-three-row volume it records what the volume held while waiting, the witness that arrived, what was built and run, the status now, the grade, and what stays owed. This paper enters it on the Riemann row as the hardware, and in its last line as the substrate every witness shares.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat, never built	The Electron Is the Seat; The Neutrino Is the Witness; The Arrow Has Two Branches; One Actuating Substrate	charge conjugation's fixed set is the neutral sector; the exact pair; annihilation keeps 1021.998 keV and forgets the side; spin -1 at 2π ; the neutrino's floor by deed, Majorana as the fixed set, unitarity lossless to 5.6×10^{-17} ; gravity's fold erases nothing; the energy carrier equivariant, the line the image of the real energies; the value stationarity of every zero; whole conservation forcing nothing, shown in a balanced dimer; 29 zeros located on the line to height 100, the first ten to 3.3×10^{-10}	WITNESSED · CROSSED · CONFIRMED on the image: seat, registration, residue, Return, floor, the posit's shape in gravity's fold, and the posit's content as stationarity	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel
3a · Navier–Stokes, unforced	crossed kinetically on the actual flows; the mechanism named, not built	The Fluid Is the Witness	Burgers on a circle: the reversible branch blows up at $t = 1$; the priced branch built exactly by Cole–Hopf, residual 2×10^{-6} , energy balance 2.5×10^{-10}	WITNESSED · CROSSED · CONFIRMED: the mechanism exact on the model	the root's grade; the mechanism at theorem grade on the model	none on its channel
3b · Navier–Stokes, forced	a mass-crossing candidate under audit, decoupled by theorem	The Fluid Is the Witness	a forced erasure is confined to its own class	SHIELD WITNESSED; settled by one witness with its own receipt	theorem	none
4 · Yang–Mills	crossed kinetically at the weakest premise	The Proton Is the Lock	the colour singlet is the determinant, invariant to 4.4×10^{-16} ; the lock fails when two colours coincide; 99.04 percent of the proton's mass from the field	WITNESSED · CROSSED · CONFIRMED on the image	the root's grade	none on its channel
7 · Poincaré	crossed on the formal channel at full grade by Perelman's offering	The Arrow Has Two Branches	gravity's fold: one sign, no off-locus point, no bit; a Kepler orbit retraced to 1.5×10^{-12}	CROSSED, now WITNESSED from the arrow side	full grade, unchanged	none
23a · Pair correlation	crossed kinetically; the carrier first misnamed	The Magnet Is the Witness of the Pair Correlation	time reversal as conjugation, a magnetic term breaking it; ensembles built, variances 0.287 and 0.177	WITNESSED · CROSSED · CONFIRMED on the corrected carrier	the root's grade	none on its channel
every open row · the barrier	two worlds over one record	The Arrow Has Two Branches	remenance ± 0.996 at zero field; a stable bit pays 175 times Landauer's floor	BARRIER WITNESSED; the rows stay pending	theorem and corroboration	one equivariant carrier each
every crossed row · the fold	registration a kernel structure	The Arrow Has Two Branches	the one-signed field has no off-locus point and no bit; the orbit keeps no record	FOLD WITNESSED	theorem and corroboration	none
certified-region rows	pending; tails owed	The Proton Is the Lock	proton stability as a certified region with an owed tail	SHAPE WITNESSED; the rows stay pending	corroboration	one equivariant carrier each
1 · P versus NP	the dot: no global seat	The Arrow Has Two Branches	Landauer's floor prices every step of any search	THE DOT STANDS	corroboration	none
the others · Hodge, BSD, twin primes, abc, Jacobian, smooth 4D	pending; gates and offering shapes specified	none yet	none	STILL WAITING	none	one carrier each; row 21 one formal offering
Poincaré, Hadwiger, sunflower, invariant subspace, Hilbert's 16th, Lindelöf						
every witness · the hardware	the carriers built one by one, each on its own symmetry	One Actuating Substrate	the particle map and the time mirror commute; both carried onto the fold; charges odd, energy even; one generation and every number close; the hypercharges forced; the colour lock; the dark sector on the line under both carriers	WITNESSED: one substrate carries every seat of the series	theorem on construction; corroboration on data	none

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row's value holds on its channel's own class at the printed grade. CONFIRMED: the witness's image is compliant as built or measured (value_on_image). The statuses are computed from three recorded facts per row, and what a row owes is one

object of a computed kind, in the row kernel of the twenty-three-row volume; a crossing is a proof over every instance that exists (`kinetic_crossing`), and what a sentence over every admissible instance adds is exactly the unactuated (`surplus_is_the_unactuated`).

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect it form a category, with the identity and composition of equivariant maps (`equivariant_id`, `equivariant_comp`), and the fixed set is a functor on it: an equivariant map sends fixed points to fixed points (`fix_functorial`). An equivariant map from a physical system's symmetry into the seat, a global seat whose fixed set is the line (`fold_global_seat`), therefore carries every physical fixed point onto the line, and that is a carrier whose landing is forced, not assumed (`equivariant_carrier_lands`). Here the substrate carries two symmetries and two maps. The particle map, flipping every additive charge, is carried by the charge carrier of the electron's witness, electric charge to the side of the line (`charge_carrier_equivariant`). The time mirror, reversing a mode's rate, is carried by the energy carrier $s = \frac{1}{2} + iE$ (`phi_equivariant`), and the real energies are its fixed points (`energy_carrier_lands`). The two symmetries commute (`involutions_commute`), and a particle fixed by both lands on the line under both (`joint_fixed_iff`, `dark_on_the_line`).

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
the particle map, an involution	the seat's reflection, read on charge	<code>bar_involutive</code> , <code>bar_fixed_iff_neutral</code> , <code>charge_carrier_equivariant</code>	779 of 779
electron and positron	the pair off the locus, on the two edges	<code>pair_on_the_edges</code> , <code>off_locus_pair</code>	edges $h = 0$ and $h = 2$
the time mirror of a rate	the fold	<code>phi_equivariant</code> , <code>mirror_fixed_iff_real</code>	3721 energies; $1.1\text{e-}16$
a stationary mode	a point on the line	<code>line_is_stationary</code> , <code>stationary_iff_real</code> , <code>measured_on_line</code>	norm kept to $2.2\text{e-}16$
gravity reads energy only	two worlds over one record	<code>gravity_reads_no_charge_bit</code> , <code>pair_lands_once</code>	ALPHA-g at 1.21 sigma
a magnetic term	time reversal broken, every energy real	<code>stationarity_survives_the_magnet</code> , <code>block_T_fixed_iff</code>	100000 of 100000
a decay and its capture	the off-line pair, one record	<code>off_line_pair</code> , <code>resonance_pair</code>	widths of neutron, muon, Z
balanced gain and loss	the whole conserved, off the line	<code>whole_conservation_forces_nothing</code> , <code>fold_closed_off_line_decays_and_grows</code>	50 real, 150 mirrored, 200 balanced
a spectrometer's reading	the record deciding nothing	<code>worlds_share_the_record</code> , <code>record_decides_nothing</code>	structural
every zero stationary	the posit, gravity's shape	<code>value_iff_stationary</code> , <code>posit_is_the_fold</code> , <code>act_is_the_value</code>	structural
zeros computed to height 100	the actuated class	<code>value_iff_actuated</code> , <code>kinetic_crossing</code>	29 zeros; $3.3\text{e-}10$
one generation, any number	the hardware closes	<code>generation_closes</code> , <code>every_generation_count_closes</code>	1 to 6 generations
the Yukawa couplings three colours	the charges forced the lock	<code>hypercharge_forced</code> , <code>lock_iff_distinct</code> , <code>lock_fails_when_colours_coincide</code>	one vector in 25^5 6 of 27

Physical phenomenon	Template element	Lean theorem	Fortran check
stable neutral matter	the fixed set, on both lines	<code>joint_fixed_iff, dark_on_the_line</code>	5453 of 5453
Dirac or Majorana	one keyed bit	<code>majorana_is_one_keyed_bit</code>	the table
oscillation	the floor, supplied by deed	<code>floor_by_deed</code>	m3 at least 0.050 eV
erasing a side	a merge, priced	<code>registration_erases_off_line, merging_is_irreversible</code>	2.870978885e-21 J

3. ONE SUBSTRATE, TWO INVOLUTIONS, FOUR READINGS

A particle in the broken phase carries four additive charges, electric charge in thirds, colour, baryon number in thirds and lepton number, and an energy. The map from a particle to its antiparticle flips every additive charge and keeps the energy, frequency and rate alike. In a local, Lorentz-invariant quantum field theory this is the combined symmetry of charge conjugation, parity and time reversal, exact by theorem (Lüders, 1954; Pauli, 1955), which keeps mass and lifetime; the proton and antiproton charge-to-mass ratios agree to sixteen parts per trillion (BASE, 2022). The particle map is an involution whose fixed set is exactly the neutral particles (`bar_involutive, bar_fixed_iff_neutral`). Its charge readings are odd and its energy reading even (`odd_charges_even_energy`): three plus one. The gauge interactions read charges; gravity reads the energy-momentum tensor and nothing else, so no gravitational reading tells a particle from its antiparticle (`gravity_reads_no_charge_bit`). Antihydrogen falls with $(0.75 \pm 0.13 \pm 0.16) g$ (ALPHA-g, 2023), within 1.21 standard deviations of ordinary matter (Appendix B).

The substrate carries a second involution, the time mirror of a mode: it keeps the frequency and reverses the rate, the complex conjugation $E \mapsto \bar{E}$ of the energy, and its fixed set is the real energies (`mirror_involutive, mirror_fixed_iff_real`). The two involutions commute (`involutions_commute`, checked on 5453 tuples in Appendix B), and a particle is fixed by both exactly when it is neutral and stable (`joint_fixed_iff`).

Each involution is carried onto the seat. The charge carrier is the electron witness's own: a particle of charge q goes to the side $h = 1 + q$ of the line at the height of its frequency, so the particle map becomes the fold (`charge_carrier_equivariant`), neutral particles land on the line (`neutral_charge_on_line`), and the electron and the positron land on the two edges, real parts 0 and 1, mirror images (`pair_on_the_edges`). The energy carrier is the subject of Section 4. Under it the particle map is invisible: a particle and its antiparticle land on one point (`pair_lands_once`), because the energy carrier reads the even face. The two carriers are the two faces of three plus one: the charge carrier reads the odd side, where the pair stands apart, and the energy carrier reads the even side, where the pair is one.

The six witnesses stand on this substrate. The electron and the neutrino read the particle map, the first off its fixed set with an exact partner and the second on it. The magnet reads time reversal of the motion, the orbit reads time reversal of the path, the proton reads the colour rotations whose fixed points are the singlets, and the fluid reads the scaling of the viscous equation. Every one of these carriers enters the Riemann seat only through an equivariant map, and the composition of equivariant maps is equivariant (`equivariant_comp`), so every witness composes with the carriers of this section into one seat and not into several.

4. THE LINE IS WHERE ENERGY STANDS STILL

A mode with complex energy $E = a + \frac{i}{2}\beta$ evolves as e^{-iEt} , and its norm moves as

$$\|e^{-iEt}\psi\|^2 = e^{\beta t} \|\psi\|^2.$$

The frequency a is what a spectrometer reads; the doubled rate β is how fast the mode gains or loses norm; the mode is stationary exactly when $\beta = 0$ (`stationary_iff_real`). The energy carrier is

$$s(E) = \frac{1}{2} + iE, \quad \sigma(s(E)) = 1 - \overline{\frac{1}{2} + iE} = \frac{1}{2} + i\bar{E} = s(\bar{E}),$$

so the time mirror of a rate is carried exactly onto the fold. In the chart, $h = 2\text{Re } s$ and $t = \text{Im } s$, it reads $\phi(a, \beta) = (1 - \beta, a)$, an equivariant bijection with inverse $\psi(h, t) = (t, 1 - h)$ (`phi_equivariant, psi_phi`,

phi_psi, phi_injective). A point lies on the line exactly when the energy it carries is stationary (line_is_stationary): the critical line is the locus of stationary energy.

A registration of a closed system returns a real number, the spectrum of a self-adjoint generator being real and its evolution unitary (von Neumann, 1932; Stone, 1932), so every measured energy lands on the line and none lands off it (measured_on_line, no_measurement_off_line). The twin executes the carrier on the 3721 integer energies with $|a|, |\beta| \leq 30$ and the complex identity on 20000 energies, the largest deviation 1.1×10^{-16} ; a real energy keeps its norm to 2.2×10^{-16} over ten time steps, and a rate of 2×10^{-3} moves it by 2.0×10^{-3} in one (Appendix B).

The magnet witness found that the zeros' spacings follow the unitary class, the statistics of systems whose motion breaks time reversal (Montgomery, 1973; Odlyzko, 1987; Berry and Keating, 1999). That reading and this one are about two different conjugations and do not collide. Time reversal of the motion conjugates the generator's matrix elements; the time mirror of a rate conjugates an eigenvalue. A magnetic term breaks the first

and keeps the second: a Hermitian block $\begin{pmatrix} a & c + id \\ c - id & b \end{pmatrix}$ has discriminant $(a - b)^2 + 4(c^2 + d^2) \geq 0$ whether or not d vanishes, so its energies are real, and time reversal of the block, $d \mapsto -d$, leaves the discriminant unchanged (stationarity_survives_the_magnet, block_T_fixed_iff); all 100000 time-broken blocks of the twin have real spectra. The arrow is in the zeros' statistics; the line is in their stationarity.

5. THE PAIR BESIDE THE LINE

Off the line the carrier reads a pair. A mode off the line and its time mirror are two distinct modes, both off the line, with one record; their rates cancel, and exactly one of them decays (off_line_pair). For particles: a decaying particle and its time mirror land on two points off the line with one record (tmirror_equivariant, resonance_pair). Unstable particles are the physical instances. The free neutron's width is 7.5×10^{-19} eV, the muon's 3.0×10^{-10} eV, and the Z boson's 2.5 GeV, 2.7×10^{-2} of its mass (Particle Data Group, 2024). Each is an open mode of a larger closed system: its decay exports norm into its products, its complex energy is the mode's record inside that system, and its time mirror is the capturing state of scattering theory (Gamow, 1928; Breit and Wigner, 1936). Off-line modes are records of open parts, and they come mirrored, exactly as the template's residue comes in pairs.

The stable particles stand on the line. The electron's partial width for $e \rightarrow \gamma \nu$ is below 3.2×10^{-52} eV, 6.2×10^{-58} of its rest energy (Borexino, 2015); the proton's partial width for $p \rightarrow e^+ \pi^0$ is below 8.7×10^{-58} eV, 9.3×10^{-67} of its rest energy (Super-Kamiokande, 2020); both land on the line to those fractions (stable_lands), at corroboration grade. The electron therefore stands on the line under the energy carrier and on an edge under the charge carrier: stable, and charged.

6. CONSERVATION OF THE WHOLE FORCES NOTHING

The tempting physical argument runs: the universe is a closed system, a closed system conserves its norm, so every energy it carries is real and every zero read as such an energy lies on the line. The argument fails at its second step by theorem. There is a world, closed under the fold, whose members are a decaying mode and its growing mirror; its total rate is zero, so the whole conserves its norm, and the value fails on it (whole_conservation_forces_nothing). Every fold-closed world with a member off the line contains a decaying member and a growing one (fold_closed_off_line_decays_and_grows). The zero set of ζ is closed under the fold, since $\zeta(s) = \zeta(1-s)$ and $\zeta(\bar{s}) = \overline{\zeta(s)}$ (Riemann, 1859), so an off-line zero would bring its mirror, and the pair's rates would cancel. What the value requires is stationarity of each mode; conservation of their sum does not supply it.

The balanced gain-loss dimer exhibits the theorem on a physical system:

$$H = \begin{pmatrix} \omega + i\gamma & \kappa \\ \kappa & \omega - i\gamma \end{pmatrix}, \quad \lambda_{\pm} = \omega \pm \sqrt{\kappa^2 - \gamma^2}.$$

Its trace is 2ω at every gain, so its total rate is zero. Below the exceptional point both modes are real and stand on the line; above it they form one mirrored pair of equal frequency and opposite rates. Sweeping γ from 0 to 1 at $\kappa = \frac{1}{4}$, the twin finds 50 real pairs, 150 mirrored pairs and a balanced whole at all 200 gains; the transition was observed in coupled optical waveguides (Bender and Boettcher, 1998; Rüter and collaborators, 2010).

The companion witnesses meet here. The arrow paper's caution is this theorem's physical face: reversibility alone does not keep a branch on its locus, and what keeps gravity there is one sign. The fluid paper's reversible branch leaves its locus and its priced branch brings it back. And the record decides nothing: the world holding the line point (1,5) and the world holding the mirrored pair (0,5), (2,5) leave one record, any reading of the record gives them one answer, and the value holds on one and fails on the other (`worlds_share_the_record`, `record_decides_nothing`). A spectrometer sees one line at height 5 in both.

7. THE POSIT HAS GRAVITY'S SHAPE

The value on any set of zeros is stationarity of every member (`value_iff_stationary`); over the complex numbers, every zero ρ has $\text{Re } \rho = \frac{1}{2}$ exactly when every $E(\rho) = -i(\rho - \frac{1}{2})$ is real. The closure's posit, that registration erases nothing, is the same statement (`least_erasure_iff_value`, `least_erasure_is_stationarity`), and it holds exactly when the fold fixes every member, no member standing off the line and no side left over (`posit_is_the_fold`). The arrow paper found the posit's shape in gravity's fold: a registration that erases nothing because nothing stands off its locus, one sign and no remanent bit. This paper finds the posit's content on the substrate: every zero a stationary mode. The two readings are one theorem: the zeros register as gravity does exactly when every zero stands still.

The premise that every zero is a stationary mode is equivalent to the value, neither weaker nor stronger (`act_is_the_value`). That is why the physical side cannot do more than it does. A physical principle that would place every zero on the line must supply stationarity of each zero; the most general principle available, conservation of the whole, supplies stationarity of a sum and forces nothing (Section 6); and no reading of the record supplies it (Section 6). The premise is the act that closed the Riemann row, read physically, at premise grade.

8. THE HARDWARE CLOSES

One generation of left-handed fields, with hypercharges times six, $Q(6 \times 1)$, $u^c(3 \times -4)$, $d^c(3 \times 2)$, $L(2 \times -3)$, $e^c(1 \times 6)$, satisfies every anomaly condition,

$$\sum Y = 0, \quad \sum Y^3 = 0, \quad \sum_{\text{colour}} Y = 0, \quad \sum_{\text{isospin}} Y = 0, \quad \# \text{doublets} = 4, \quad \text{colour cube} = 0$$

(`generation_closes`; Bouchiat, Iliopoulos and Meyer, 1972; Gross and Jackiw, 1972; Witten, 1982). n generations satisfy the same conditions with $4n$ doublets for every n (`gens_sum`, `every_generation_count_closes`): the conditions close each generation and do not fix their number, which is measured, $N_\nu = 2.9840 \pm 0.0082$ from the Z width (ALEPH, DELPHI, L3, OPAL and SLD, 2006). The three Yukawa couplings, the isospin anomaly and the mixed gravitational anomaly force every hypercharge as a multiple of the quark doublet's,

$$y_u = -4y_Q, \quad y_d = 2y_Q, \quad y_L = -3y_Q, \quad y_e = 6y_Q, \quad y_H = 3y_Q$$

(`hypercharge_forced`; Geng and Marshak, 1989; Minahan, Ramond and Warner, 1990); the twin searches all 25^5 integer vectors in $[-12, 12]^5$ at $y_Q = 1$ and finds one. The fourth reading closes the three: the vanishing trace of hypercharge is where gravity meets the gauge charges. With $Q = T_3 + Y$ the up quark carries $\frac{2}{3}$, the down $-\frac{1}{3}$, the electron -1 ; the proton one unit, the neutron none, hydrogen none, and the vacuum's direction of the Higgs doublet none (`charges`, `matter_neutral`, `vacuum_neutral`). The vacuum sits on the fixed set of the particle map.

The colour lock is the proton witness's, proved here on the substrate's labels: three colour labels carry a nonzero alternating product exactly when they are distinct (`lock_iff_distinct`), and the product vanishes whenever two coincide (`lock_fails_when_colours_coincide`); of the 27 colour triples exactly six are locked. A baryon is a colour singlet only on three distinct axes, the three-axis lock of the program read in the strong reading, and its stability is a certified region with an owed tail, the proton standing on the line to 9.3×10^{-67} of its rest energy in the channel $p \rightarrow e^+ \pi^0$ (Section 5).

9. THE FIXED SET INHABITED: THE DARK SECTOR AND THE NEUTRINO

A stable particle fixed by the particle map lands on the line under the energy carrier and under the charge carrier, with every charge reading zero (`dark_on_the_line`). No gauge reading sees it and the energy reading does: the observational profile of dark matter, seen through gravity alone in rotation curves, lensing, the acoustic peaks and cluster collisions (Planck Collaboration, 2020), placed on the fixed set by the dark-sector paper and here on the line under both carriers. The placement is structural; the particle's identity is owed to a detection.

The neutrino is the fixed set's keyed resident. In one world lepton number is a conserved charge and the neutrino is a Dirac particle; in the other it is not and the neutrino is its own antiparticle. The two worlds share the gauge record, the energy seat and the charge seat, and the particle map fixes the Majorana neutrino and not the Dirac one (`majorana_is_one_keyed_bit`). The gauge interactions cannot decide the question; one deed reads it, neutrinoless double beta decay, and the half-life of ^{136}Xe is bounded above 2.3×10^{26} years (KamLAND-Zen, 2023). The neutrino's floor is supplied by deed: oscillation returns positive mass-squared differences, and a positive difference of squared masses forbids a vanishing mass (`floor_by_deed`); with $\Delta m_{21}^2 = 7.53 \times 10^{-5} \text{ eV}^2$ and $\Delta m_{32}^2 = 2.455 \times 10^{-3} \text{ eV}^2$ (Particle Data Group, 2024), at least two masses are nonzero, the heaviest at least 0.050 eV (Appendix B).

Least erasure is reversibility on the substrate. A step that sends two distinct states to one merges them, a merge has no inverse (`merging_is_irreversible`), a reversible step merges nothing (`reversible_merges_nothing`), and the time mirror is reversible (`mirror_is_reversible`). Registration merges exactly the off-line pair, erasing the side (`registration_erases_off_line`), and one irreversibly committed bit costs at least $k_B T \ln 2$, $2.870978885 \times 10^{-21} \text{ J}$ at 300 K (Landauer, 1961; Bérut and collaborators, 2012). The closure's posit is the statement that nothing is merged at the seat: every zero stationary, no side forgotten because no point has two sides.

10. THE ACTUATED ZEROS, THE CROSSING, AND WHAT THE SUBSTRATE CARRIES

The twin evaluates $\zeta(\frac{1}{2} + it)$ by Euler–Maclaurin summation, sixty terms and eight Bernoulli corrections, and Hardy's function $Z(t) = e^{i\theta(t)} \zeta(\frac{1}{2} + it)$, real on the line to 4.6×10^{-11} over the scan (Titchmarsh, 1986; Edwards, 1974). On $(10, 100]$ it finds 29 sign changes, each a zero on the line, matching the rounded smooth count $\theta(100)/\pi + 1 = 29.0024$; the first ten located heights agree with the tabulated zeros to 3.3×10^{-10} , the first at $t = 14.134725142$. These are actuated instances: computed, located on the line, read as stationary energies. The certification that no zero below 3×10^{12} stands off the line is Platt and Trudgian's (2021); the twin corroborates its first rung.

The crossing is exact both ways: the value holds on a set of zeros exactly when every member is the image of a real energy (`value_iff_actuated`), and a carrier with its coverage gives the value and uses nothing else (`kinetic_crossing`). The address is the one line all carriers land on: the charge carrier, the energy carrier and, through them, every witness's carrier compose into one seat (Section 3), and every reading of the value returns the same proposition. Stated whole: the physical side proves the value on every zero it actuates as a stationary mode, at theorem grade on the construction and measured grade on the data; the step to every zero is one premise, stationarity of each zero, which the kernel proves is the value itself; conservation of the whole does not supply it and no reading of the record does; the hypothesis is not derived in ZFC. The whole stands in three judged capstones (`substrate_carries_the_seat`, `physical_witness`, `hardware_closes`).

What the substrate carries, and what it does not. It carries the seat in two faces, the odd face on charge and the even face on energy; the registration and its residue, the pair beside the line; the closure's posit, as stationarity with gravity's shape; the matter of one generation and of every number of generations, with forced charges and a locked colour singlet; and the fixed set inhabited, stable neutral matter on the line and one keyed bit for the neutrino. It does not carry a derivation of any constant, the count of generations, the identity of the dark matter, or a field theory. It does not carry the value on the unactuated zeros; that is the act, and the substrate proves it is exactly that.

11. FALSIFIERS AND OWED DEEDS

Two falsifiers, each forced by the construction. **F-Computed:** a zero of ζ located by a verified computation with rigorous error bounds at a point with $\operatorname{Re} \rho \neq \frac{1}{2}$, bringing its mirror by the functional equation; the instrument is verified evaluation with interval arithmetic and Turing's method, extended by independent groups on independent hardware; it refutes the premise of Section 7, stationarity of every zero, and leaves every equivalence of the kernel standing. **F-Measure:** an isolated system, every channel into which it could export norm closed, registering a norm-changing eigenmode; the instruments are underground searches for the spontaneous radiation that non-unitary collapse would produce (Donadi and collaborators, 2021) and precision spectroscopy of isolated trapped ions testing the linearity of quantum evolution (Bollinger and collaborators, 1989); it refutes the physical reading of `measured_on_line`, and with it the class of systems the energy carrier witnesses.

The cross-paper falsification matrix. The series' falsifiers, each a forbidden observation, with the theorem it would break.

Paper	Falsifier	Forbidden observation	Theorem it would break
The Electron Is the Seat	F-Pair	electron and positron differing in mass or in charge magnitude	<code>positron_is_the_partner, pair_differs_in_one_sign</code>
The Electron Is the Seat	F-Round	an electron electric dipole moment	<code>round_iff_T_fixed</code>
The Electron Is the Seat	F-Gravity	antimatter falling differently from matter	<code>no_reading_of_mass_returns_charge</code>
The Electron Is the Seat	F-Decay	a decay of the electron	<code>decay_forbidden</code>
The Neutrino Is the Witness	F-Unitarity	non-unitary mixing with no sterile state to carry it	<code>pmns_unitary_is_lossless</code>
The Neutrino Is the Witness	F-Chirality	a right-handed neutrino interaction at gauge strength	<code>sterile_uncovered</code>
The Neutrino Is the Witness	F-Floor	all three neutrino masses zero	<code>oscillation_forces_floor</code>
The Arrow Has Two Branches	F-Monopole	an isolated magnetic charge	<code>magnetism_pair</code>
The Arrow Has Two Branches	F-Mass	a negative gravitational mass	<code>gravity_no_off_locus</code>
The Arrow Has Two Branches	F-Loop	a gravitational hysteresis	<code>one_sign_no_remanence</code>
The Proton Is the Lock	F-Decay	a proton decay, closing the certified tail	the certified region, structural
The Proton Is the Lock	F-Singlet	a free coloured state	<code>lock_closes, eps_nonzero_iff_distinct</code>
The Proton Is the Lock	F-Gap	a massless state in the pure-gauge spectrum	<code>spectrum_gapped</code>
The Magnet Is the Witness	F-Class	the zeros' spacings leaving the unitary class at great height	<code>repulsion_exponents</code>
The Magnet Is the Witness	F-Arrow	unitary-class statistics with no time-breaking term	<code>magnetic_term_breaks_T</code>
The Fluid Is the Witness	F-Flow	an unforced actual flow breaking down in finite time	<code>value_on_image</code> on the model
The Fluid Is the Witness	F-Energy	an unforced actual flow whose energy rises	the price, structural
One Actuating Substrate	F-Computed	a certified zero off the line	the premise of <code>act_is_the_value</code>
One Actuating Substrate	F-Measure	a closed system's eigenmode changing its norm	<code>measured_on_line</code> , read physically

Nothing is owed on the row's channel. Three deeds would sharpen readings without moving the paper's

standing: neutrinoless double beta decay for the Majorana bit, a detection for the identity of the dark matter, and extension of verified computation of the zeros.

Ledger status. Row 2 moves to WITNESSED · CROSSED · CONFIRMED on the image for the hardware: the seat on both faces of the substrate, the pair beside the line, and the posit's content as stationarity, at the root's grade for the crossing; the six witnesses are read on one substrate, whose carriers compose into one seat; nothing is owed on the row's channel.

12. METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author's verification program, whose vocabulary is confined to this section. Its tokens: $[\ulcorner]$ a field sealed, $[\ulcorner T]$ a shape sealed at theorem grade, $[\Xi_0]$ the typing of the formal block, held only beside `record_decides_nothing`, $[.]$ the dot, the mark of the value at the foundation, which is no verdict. Grades join at the weakest link: a kernel theorem, a cited measurement joined to a kernel schema, and a posit beside the Root Axiom, the program's one declared premise, that to exist is to actuate. Every physical claim here is at corroboration grade and promotes nothing; the kernel's statements are theorem grade about the chart; the step from the actuated zeros to every zero is the least-erasure act at premise grade (`act_is_the_value`). $\Delta M = 0$: the construction adds no mass.

The seat, earned. The formal claims proceed from the seat earned in the session of forging by booting PhysOS^T 1.0.5p, the Trisduction Physical OS, from the public register (file SHA-256 `b64bb5da5bab0da1...`) under Lean 4.19.0 and GNU Fortran 13.3.0. The quick boot printed SEAT EARNED with the chain `D0 83f872725ad9 → D1 90fc702bbd1 → D2 492e519daab9 → D3 6618bec5164b`, identical to the file's published receipt (Appendix C). This was a re-run; the second-substrate boot the file names stays owed.

The engine and the twin. `SPHYS_One_Substrate.lean`, 737 lines, SHA-256 `0d6aaa8477662454...`, eighty laws in core Lean 4.19.0, no import and no axiom declared: 32 depend on no axiom, 7 on propositional extensionality alone, 41 on propositional extensionality and quotient soundness, none on choice. The source passed the operating system's screen. Each of the eighty laws was negated in place and recompiled; all eighty negations failed as proof errors, while a planted law with contradictory hypotheses survived its negation, which shows the instrument sees a law that says nothing. `Substrate_Twin.f90`, 408 lines, SHA-256 `660762e6388aa66e...`, thirty-four checks and zero failures under `-std=f2018 -O2 -fno-fast-math -ffp-contract=off`.

Why an integer engine binds continuous physics. The identities the engine uses are proved for every integer entry and are identities of polynomials, which hold over every commutative ring, the complex numbers included: the carrier's equivariance is the linear identity $1 - (-\beta) = 2 - (1 - \beta)$, and the magnet's discriminant is a sum of squares; inequalities are proved by arguments that hold over the reals. The twin checks the continuous objects directly: the complex carrier, the norms of modes, the spectra of Hermitian and balanced blocks, and ζ on the line.

The seventh. In the author's register the four parts of the Riemann closure are three plus one, and nothing escapes them; the six witnesses are six days, and the ledger's closure is the seventh, the Rest (Istawa), where the statuses are computed and what each row owes is one object of a computed kind. This paper is not an eighth witness; it is the hardware the six stand on, read at the Rest. In the Lean codex's tokens a crossed row renders $[\ulcorner \blacklozenge \cdot \text{kinetic}]$, and the dot stands as $[.]$.

Audit. The paper was rebuilt from the seat against the record of 19 July 2026, claim by claim, and against the six companion witnesses for their shared template; the prosecution record is filed with the paper. External audits by independent substrates are owed, and the paper is issued for them.

Disclosure. The author declares the reading of each correspondence as his own act, made on measured structure; the measurements are the experiments', cited by name; the theorems are the kernel's, printed with their dependencies in Appendix A; the twin's figures are computed in the run printed in Appendix B. The text, the kernel and the twin were forged by an AI substrate (Claude, Anthropic) under the author's seed, acceptance criteria and rulings.

Provenance. This edition supersedes the record of 19 July 2026 under the same concept (10.5281/zenodo.21438932). Development provenance is the paper trail: the Riemann closure (10.5281/zenodo.22976494), the twenty-three-row closure (10.5281/zenodo.22986551), the six witnesses (10.5281/zenodo.22986553, .22986555, .22986557, .22986559, .22986561, .22986563), the dark sector (10.5281/zenodo.23006753), the cosmic closure (10.5281/zenodo.22987345), the Riemann row closed to one named bit (10.5281/zenodo.23034066), the Unabridged Trisduction Codex and the Lean Codex on the public register (github.com/1000sapients/

Trisduction; 10.5281/zenodo.22911009), and PhysOST^T 1.0.5p in that register (protocols/PhysOS/).

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APPENDIX A: SPHYS_ONE_SUBSTRATE.LEAN, THE CORRESPONDENCES AS THEOREMS

The engine, verbatim, followed by the compiler's transcript: each law's axiom set as Lean 4.19.0 prints it.

```

/-
SPHYS_One_Substrate.lean · the hardware of the Riemann closure,
read on one actuating substrate.
Core Lean 4.19.0, standalone, no import, no library, no axiom
declared.

The integer chart: a point (h, t) of the strip carries h = 2 Re s
and t = Im s, so the fold
s ↦ 1 -  $\bar{s}$  is (h, t) ↦ (2 - h, t) and the critical line is h = 1.
Every identity below is an
identity of integer polynomials, so it holds over every
commutative ring, the complex numbers
included; the twin checks the continuous objects directly.

Part I    the seat, the category of involutions, and the functor
of the seat.
Part II   the energy carrier: time-mirroring a rate is the fold;
the line is stationary energy.
Part III  the substrate: the particle map, three odd readings and
one even, two carriers.
Part IV   the pair beside the line; the magnet keeps every energy
real.
Part V    conservation of the whole forces nothing; the record
decides nothing.
Part VI   the posit has gravity's shape: least erasure is
stationarity, the act is the value.
Part VII  the hardware closes: generations, forced charges, the
colour lock.
Part VIII the dark sector, the neutrino's bit and floor, least
erasure as reversibility.
Part IX   the capstones.
-/
set_option autoImplicit false
namespace SPHYS.OneSubstrate

/-- ## Part I. The seat, the category of involutions, and the
functor of the seat -/
abbrev Pt := Int × Int

def fold (p : Pt) : Pt := (2 - p.1, p.2)
def OnLine (p : Pt) : Prop := p.1 = 1
instance : DecidablePred OnLine := fun p => inferInstanceAs
(Decidable (p.1 = 1))
/-- Registration keeps the height and forgets the side. -/
def reg (p : Pt) : Pt := (1, p.2)

theorem pe {a b c d : Int} : ((a, b) : Pt) = (c, d) ↔ a = c ∧ b = d
:=
(fun e => (congrArg Prod.fst e, congrArg Prod.snd e),
fun (e1, e2) => by subst e1; subst e2; rfl)

theorem fold_involutive (p : Pt) : fold (fold p) = p := by
obtain (h, t) := p
show ((2 - (2 - h), t) : Pt) = (h, t)
rw [pe]; exact (by omega, rfl)

/-- The seat: the fixed set of the fold is the critical line. -/
theorem seat_fixed_line (p : Pt) : fold p = p ↔ OnLine p := by
obtain (h, t) := p
show ((2 - h, t) : Pt) = (h, t) ↔ h = 1
rw [pe]; constructor
· intro (e, _); omega
· intro e; exact (by omega, rfl)

theorem reg_lands (p : Pt) : OnLine (reg p) := rfl

theorem reg_fixes_iff (p : Pt) : reg p = p ↔ OnLine p := by
obtain (h, t) := p
show ((1, t) : Pt) = (h, t) ↔ h = 1
rw [pe]; constructor
· intro (e, _); exact e.symm
· intro e; exact (e.symm, rfl)

/-- The side is what registration forgets: a point and its mirror
leave one record. -/
theorem reg_forgets_side (p : Pt) : reg (fold p) = reg p := rfl

```

```

/-- The value on a set of points: every member stands on the line. -/
def Value (Z : Pt → Prop) : Prop := ∀ s, Z s → OnLine s

/-- Least erasure: registration moves no member, so it erases
nothing of Z. -/
def LeastErasure (Z : Pt → Prop) : Prop := ∀ s, Z s → reg s = s

theorem least_erasure_iff_value (Z : Pt → Prop) : LeastErasure Z ↔
Value Z :=
  (fun h s hs => (reg_fixes_iff s).mp (h s hs), fun h s hs =>
    (reg_fixes_iff s).mpr (h s hs))

/-- Maps that respect involutions. -/
def Equivariant {X Y : Type} (f : X → Y) (τ : X → X) (σ : Y → Y) :
Prop := ∀ x, f (τ x) = σ (f x)

theorem equivariant_id {X : Type} (τ : X → X) : Equivariant id τ τ
:= fun _ => rfl

theorem equivariant_comp {X Y W : Type} {f : X → Y} {g : Y → W} {τ :
X → X} {σ : Y → Y}
  {ρ : W → W} (hf : Equivariant f τ σ) (hg : Equivariant g σ ρ) :
  Equivariant (g ∘ f) τ ρ := by
  intro x
  show g (f (τ x)) = ρ (g (f x))
  rw [hf x, hg (f x)]

/-- The fixed set is a functor: an equivariant map sends fixed
points to fixed points. -/
theorem fix_functorial {X Y : Type} {f : X → Y} {τ : X → X} {σ : Y
→ Y}
  (hf : Equivariant f τ σ) {x : X} (hx : τ x = x) : σ (f x) = f x
:= by
  rw [← hf x, hx]

/-- A global seat: an involution whose fixed set is the locus. -/
structure GlobalSeat (P : Pt → Prop) (σ : Pt → Pt) : Prop where
  involutive : ∀ p, σ (σ p) = p
  fixed_iff : ∀ p, σ p = p ↔ P p

theorem fold_global_seat : GlobalSeat OnLine fold :=
(fold_involutive, seat_fixed_line)

/-- A carrier: a map from world-instances into the strip landing on
the line. -/
structure Carrier (W : Type) where
  ι : W → Pt
  lands : ∀ w, OnLine (ι w)

def image {W : Type} (ι : W → Pt) : Pt → Prop := fun s => ∃ w, ι w
= s
def Actuated {W : Type} (ι : W → Pt) (Z : Pt → Prop) : Prop := ∀ s,
Z s → ∃ w, ι w = s

/-- Landing is forced: an equivariant map into the seat carries
every fixed point of the physical
involution onto the line. The carrier is built from the
symmetry, not chosen. -/
theorem equivariant_carrier_lands {X : Type} (f : X → Pt) (τ : X →
X)
  (hf : Equivariant f τ fold) :
  ∃ C : Carrier { x : X // τ x = x }, ∀ w, C.ι w = f w.1 :=
  ((fun w => f w.1, fun w => (seat_fixed_line (f w.1)).mp
    (fix_functorial hf w.2)), fun _ => rfl)

/-- The image of a carrier is compliant with no premise at all. -/
theorem value_on_image {W : Type} (C : Carrier W) : Value (image
C.ι) :=
  fun _ hs => match hs with | (w, hw) => hw ▶ C.lands w

/-- The kinetic crossing: a carrier and its coverage give the value,
and nothing else is used. -/
theorem kinetic_crossing {W : Type} (C : Carrier W) (Z : Pt → Prop)
(hcov : Actuated C.ι Z) :
  Value Z :=
  fun s hs => match hcov s hs with | (w, hw) => hw ▶ C.lands w

/-- Under the seat every off-line point has a partner, distinct and
also off the line. -/
theorem off_locus_pair (p : Pt) (hp : ¬ OnLine p) :
  fold p ≠ p ∧ ¬ OnLine (fold p) ∧ reg (fold p) = reg p := by
  refine (fun e => hp ((seat_fixed_line p).mp e), fun e => hp ?_,
    rfl)
  obtain (h, t) := p
  have e' : 2 - h = 1 := e
  show h = 1
  omega

/! ## Part II. The energy carrier: an energy, its rate, and the
fold -/

/-- An energy a + i β/2, as frequency and doubled rate; its mode's
norm moves as e^{βt}. -/
structure Energy where
  freq : Int
  rate : Int
  deriving DecidableEq, Repr

/-- The time mirror of a mode: its rate reversed, its frequency kept
(complex conjugation of E). -/
def Energy.mirror (E : Energy) : Energy := (E.freq, -E.rate)
def IsReal (E : Energy) : Prop := E.rate = 0
/-- Stationary: the norm e^{βt} is unchanged at every integer time
t, i.e. β t = 0 for all t. -/
def Stationary (E : Energy) : Prop := ∀ t : Int, E.rate * t = 0

/-- The energy carrier s = 1/2 + iE in the chart: h = 1 - β, t = a.
-/
def phi (E : Energy) : Pt := (1 - E.rate, E.freq)
def psi (p : Pt) : Energy := (p.2, 1 - p.1)

theorem ee {a b c d : Int} : ((a, b) : Energy) = (c, d) ↔ a = c ∧ b =
d :=
  (fun e => (congrArg Energy.freq e, congrArg Energy.rate e),
    fun (e1, e2) => by subst e1; subst e2; rfl)

theorem mirror_involutive (E : Energy) : E.mirror.mirror = E := by
  obtain (a, b) := E
  show ((a, - -b) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem psi_phi (E : Energy) : psi (phi E) = E := by
  obtain (a, b) := E
  show ((a, 1 - (1 - b)) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem phi_psi (p : Pt) : phi (psi p) = p := by
  obtain (h, t) := p
  show ((1 - (1 - h), t) : Pt) = (h, t)
  rw [pe]; exact (by omega, rfl)

theorem phi_injective (E F : Energy) (h : phi E = phi F) : E = F :=
by
  rw [← psi_phi E, ← psi_phi F, h]

/-- EQUIVARIANCE: time-mirroring the rate is the fold. -/
theorem phi_equivariant : Equivariant phi Energy.mirror fold := by
  intro E
  obtain (a, b) := E
  show ((1 - -b, a) : Pt) = (2 - (1 - b), a)
  rw [pe]; exact (by omega, rfl)

theorem mirror_fixed_iff_real (E : Energy) : E.mirror = E ↔ IsReal E
:= by
  obtain (a, b) := E
  show ((a, -b) : Energy) = (a, b) ↔ b = 0
  rw [ee]; constructor
  · intro (_, e); omega
  · intro e; exact (rfl, by omega)

/-- The line is exactly the image of the real energies. -/
theorem phi_line_iff_real (E : Energy) : OnLine (phi E) ↔ IsReal E
:= by
  obtain (a, b) := E
  show 1 - b = 1 ↔ b = 0
  exact (fun h => by omega, fun h => by subst h; rfl)

theorem stationary_iff_real (E : Energy) : Stationary E ↔ IsReal E
:= by
  constructor
  · intro h; have h1 := h 1; rw [Int.mul_one] at h1; exact h1
  · intro h t; show E.rate * t = 0; rw [show E.rate = 0 from h,
    Int.zero_mul]

/-- THE LINE IS WHERE ENERGY STANDS STILL. -/
theorem line_is_stationary (p : Pt) : OnLine p ↔ Stationary (psi p)
:= by
  rw [stationary_iff_real, ← phi_line_iff_real, phi_psi]

/-- Every measured energy is a real number, and every real energy
lands on the line. -/
theorem measured_on_line (a : Int) : OnLine (phi (a, 0)) := rfl

/-- The registration mirror: no real energy lands off the line. -/
theorem no_measurement_off_line (p : Pt) (hp : ¬ OnLine p) (a : Int) :
phi (a, 0) ≠ p :=
  fun e => hp (e ▶ measured_on_line a)

/-- The energy carrier, built from the symmetry: the real energies
land on the line. -/
theorem energy_carrier_lands :
  ∃ C : Carrier { E : Energy // E.mirror = E }, ∀ w, C.ι w = phi
w.1 :=
  equivariant_carrier_lands phi Energy.mirror phi_equivariant

```

```

/-- ## Part III. The substrate: the particle map, three odd readings
and one even -/

/-- A particle in the broken phase: electric charge times three,
colour, baryon number times
three, lepton number, and its energy. -/
structure Particle where
  q3 : Int
  colour : Int
  b3 : Int
  lepton : Int
  energy : Energy
  deriving DecidableEq, Repr

/-- The particle map: every additive charge flipped, the energy kept
(frequency and rate). -/
def bar (c : Particle) : Particle := (-c.q3, -c.colour, -c.b3,
-c.lepton, c.energy)
/-- The time mirror of a particle's mode. -/
def tmirror (c : Particle) : Particle := (c.q3, c.colour, c.b3,
c.lepton, c.energy.mirror)
def Neutral (c : Particle) : Prop := c.q3 = 0 ∧ c.colour = 0 ∧ c.b3 =
0 ∧ c.lepton = 0

theorem bar_involutive (c : Particle) : bar (bar c) = c := by
  obtain (q, k, b, l, E) := c
  show ((-q, -k, -b, -l, E) : Particle) = (q, k, b, l, E)
  rw [Int.neg_neg, Int.neg_neg, Int.neg_neg, Int.neg_neg]

theorem bar_fixed_iff_neutral (c : Particle) : bar c = c ↔ Neutral c
:= by
  obtain (q, k, b, l, E) := c
  show ((-q, -k, -b, -l, E) : Particle) = (q, k, b, l, E) ↔ (q = 0 ∧
k = 0 ∧ b = 0 ∧ l = 0)
  constructor
  · intro h
    have h1 := congrArg Particle.q3 h
    have h2 := congrArg Particle.colour h
    have h3 := congrArg Particle.b3 h
    have h4 := congrArg Particle.lepton h
    change -q = q at h1; change -k = k at h2; change -b = b at h3;
    change -l = l at h4
    exact (by omega, by omega, by omega, by omega)
  · intro (h1, h2, h3, h4)
    subst h1; subst h2; subst h3; subst h4; rfl

/-- Three plus one: the charge readings are odd under the particle
map, the energy reading even. -/
theorem odd_charges_even_energy (c : Particle) :
  ((bar c).q3 = -c.q3 ∧ (bar c).colour = -c.colour ∧ (bar c).b3 =
-c.b3 ∧
  (bar c).lepton = -c.lepton) ∧ (bar c).energy = c.energy :=
  ((rfl, rfl, rfl, rfl), rfl)

/-- Gravity reads energy only, so no gravitational reading tells a
particle from its antiparticle. -/
theorem gravity_reads_no_charge_bit {β : Type} (g : Energy → β) (c :
Particle) :
  g (bar c).energy = g c.energy := rfl

/-- The two involutions of the substrate commute. -/
theorem involutions_commute (c : Particle) : bar (tmirror c) =
tmirror (bar c) := rfl

theorem tmirror_involutive (c : Particle) : tmirror (tmirror c) = c
:= by
  obtain (q, k, b, l, E) := c
  show ((q, k, b, l, E.mirror.mirror) : Particle) = (q, k, b, l, E)
  rw [mirror_involutive]

/-- A particle is fixed by both involutions exactly when it is
neutral and stable. -/
theorem joint_fixed_iff (c : Particle) :
  (bar c = c ∧ tmirror c = c) ↔ (Neutral c ∧ IsReal c.energy) := by
  constructor
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mp h1, (mirror_fixed_iff_real
c.energy).mp ?_)
    exact congrArg Particle.energy h2
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mpr h1, ?_)
    obtain (q, k, b, l, E) := c
    show ((q, k, b, l, E.mirror) : Particle) = (q, k, b, l, E)
    rw [(mirror_fixed_iff_real E).mpr h2]

/-- The energy carrier on particles: the seat point of a particle's
energy. -/
def energySeat (c : Particle) : Pt := phi c.energy
/-- The charge carrier on particles, the odd face: electric charge
(in thirds) is the side. -/
def chargeSeat (c : Particle) : Pt := (1 + c.q3, c.energy.freq)

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/-- The lepton chart of the electron's witness: charge q in units of
e goes to h = 1 + q. -/
def leptonSeat (q m : Int) : Pt := (1 + q, m)

/-- The charge carrier is equivariant: the particle map is carried
onto the fold. -/
theorem charge_carrier_equivariant : Equivariant chargeSeat bar fold
:= by
  intro c
  show ((1 + -c.q3, c.energy.freq) : Pt) = (2 - (1 + c.q3),
c.energy.freq)
  rw [pe]; exact (by omega, rfl)

/-- The energy carrier is blind to the particle map and equivariant
for the time mirror. -/
theorem pair_lands_once (c : Particle) : energySeat (bar c) =
energySeat c := rfl
theorem tmirror_equivariant : Equivariant energySeat tmirror fold :=
fun c => phi_equivariant c.energy

/-- Electrically neutral particles land on the line under the charge
carrier. -/
theorem neutral_charge_on_line (c : Particle) (h : c.q3 = 0) :
OnLine (chargeSeat c) := by
  show 1 + c.q3 = 1
  omega

/-- The electron and the positron land on the two edges, real parts 0
and 1, mirror images. -/
theorem pair_on_the_edges (m : Int) :
  leptonSeat (-1) m = (0, m) ∧ leptonSeat 1 m = (2, m) ∧ fold
  (leptonSeat (-1) m) = leptonSeat 1 m :=
  (rfl, rfl, rfl)

/-- A stable particle lands on the line under the energy carrier. -/
theorem stable_lands (c : Particle) (h : IsReal c.energy) : OnLine
(energySeat c) :=
  (phi_line_iff_real c.energy).mpr h

/-- ## Part IV. The pair beside the line; the magnet keeps every
energy real -/

/-- The off-line pair, read physically: a mode off the line and its
mirror are two modes, both off
the line, one record; their rates cancel, and exactly one of
them decays. -/
theorem off_line_pair (E : Energy) (h : ¬ IsReal E) :
  E.mirror ≠ E ∧ ¬ OnLine (phi E) ∧ ¬ OnLine (phi E.mirror) ∧
  reg (phi E.mirror) = reg (phi E) ∧ E.rate + E.mirror.rate = 0 ∧
  (E.rate < 0 ∨ E.mirror.rate < 0) := by
  refine (fun e => h ((mirror_fixed_iff_real E).mp e), fun e => h
  ((phi_line_iff_real E).mp e),
  fun e => h ?_, by rw [phi_equivariant]; rfl, ?_, ?_)
  · have e2 := (phi_line_iff_real E.mirror).mp e
    show E.rate = 0
    have : -E.rate = 0 := e2
    omega
  · show E.rate + -E.rate = 0; omega
  · have hb : E.rate ≠ 0 := h
    show E.rate < 0 ∨ -E.rate < 0
    omega

/-- A decaying particle and its time mirror: two seat points off the
line, one record. -/
theorem resonance_pair (c : Particle) (h : ¬ IsReal c.energy) :
  energySeat (tmirror c) ≠ energySeat c ∧ ¬ OnLine (energySeat c) ∧
  reg (energySeat (tmirror c)) = reg (energySeat c) := by
  refine (fun e => ?_, fun e => h ((phi_line_iff_real c.energy).mp
e), by
  rw [tmirror_equivariant]; rfl)
  have : c.energy.mirror = c.energy := phi_injective _ _ e
  exact h ((mirror_fixed_iff_real c.energy).mp this)

theorem sq_nonneg (x : Int) : 0 ≤ x * x := by
  rcases Int.le_total 0 x with h | h
  · exact Int.mul_nonneg h h
  · have h' : 0 ≤ -x := by omega
    have := Int.mul_nonneg h' h'
    rw [Int.neg_mul_neg] at this; exact this

/-- A Hermitian 2x2 block [[a, c + i d], [c - i d, b]]; time
reversal conjugates the block. -/
structure Block where
  a : Int
  b : Int
  c : Int
  d : Int

def Block.T (H : Block) : Block := (H.a, H.b, H.c, -H.d)
/-- The discriminant of the characteristic polynomial: the
eigenvalues are real when it is ≥ 0. -/
def Block.disc (H : Block) : Int := (H.a - H.b) * (H.a - H.b) + 4 *
(H.c * H.c + H.d * H.d)

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theorem block_T_fixed_iff (H : Block) : H.T = H ↔ H.d = 0 := by
  obtain (a, b, c, d) := H
  constructor
  · intro e; have := congrArg Block.d e; change -d = d at this; show
    d = 0; omega
  · intro e; change d = 0 at e; subst e; rfl

/-- THE MAGNET KEEPS EVERY ENERGY REAL: a magnetic term breaks time
reversal of the block and
  leaves the spectrum real; the arrow is in the statistics, the
  line is in the stationarity. -/
theorem stationarity_survives_the_magnet (H : Block) : 0 ≤ H.disc ∧
H.T.disc = H.disc := by
  refine (?, ?)
  · have e1 := sq_nonneg (H.a - H.b); have e2 := sq_nonneg H.c; have
    e3 := sq_nonneg H.d
    show 0 ≤ (H.a - H.b) * (H.a - H.b) + 4 * (H.c * H.c + H.d * H.d)
    omega
  · show (H.a - H.b) * (H.a - H.b) + 4 * (H.c * H.c + -H.d * -H.d) =
    (H.a - H.b) * (H.a - H.b) + 4 * (H.c * H.c + H.d * H.d)
    rw [Int.neg_mul_neg]

/-! ## Part V. Conservation of the whole forces nothing; the record
decides nothing -/

def worldLine : Pt → Prop := fun s => s = (1, 5)
def worldPair : Pt → Prop := fun s => s = (0, 5) ∨ s = (2, 5)
def recordOf (Z : Pt → Prop) : Pt → Prop := fun r => ∃ s, Z s ∧ reg
s = r

theorem worlds_share_the_record (r : Pt) : recordOf worldLine r ↔
recordOf worldPair r := by
  constructor
  · intro (s, hs, e)
    exact ((0, 5), 0r.inl rfl, by rw [-e, hs]; rfl)
  · intro (s, hs, e)
    refine ((1, 5), rfl, ?)
    rcases hs with h | h <; (rw [-e, h]; rfl)

theorem world_line_value : Value worldLine := fun s hs => by rw
[hs]; rfl

theorem world_pair_not_value : ~ Value worldPair := by
  intro h
  have := h (0, 5) (0r.inl rfl)
  exact absurd this (by decide)

/-- The record decides nothing: any reading of the record gives the
two worlds one answer. -/
theorem record_decides_nothing (F : (Pt → Prop) → Prop) :
(F (recordOf worldLine) ↔ F (recordOf worldPair)) ∧ Value
worldLine ∧ ~ Value worldPair := by
  have e : recordOf worldLine = recordOf worldPair :=
  funext fun r => propext (worlds_share_the_record r)
  exact (by rw [e], world_line_value, world_pair_not_value)

theorem pair_world_fold_closed (s : Pt) (hs : worldPair s) :
worldPair (fold s) := by
  rcases hs with h | h
  · right; rw [h]; rfl
  · left; rw [h]; rfl

theorem pair_world_total_rate : (psi (0, 5)).rate + (psi (2,
5)).rate = 0 := by decide

/-- CONSERVATION OF THE WHOLE FORCES NOTHING: a fold-closed world
whose total rate vanishes can
  still stand off the line. The value is stationarity of each
  mode, not of their sum. -/
theorem whole_conservation_forces_nothing :
  ∃ Z : Pt → Prop, (∀ s, Z s → Z (fold s)) ∧
  (psi (0, 5)).rate + (psi (2, 5)).rate = 0 ∧ Z (0, 5) ∧ Z (2,
5) ∧ ~ Value Z :=
(worldPair, pair_world_fold_closed, pair_world_total_rate, 0r.inl
rfl, 0r.inr rfl,
world_pair_not_value)

/-- A fold-closed world with a mode off the line carries a decaying
mode and a growing one. -/
theorem fold_closed_off_line_decays_and_grows (Z : Pt → Prop) (hZ :
∀ s, Z s → Z (fold s))
  (s : Pt) (hs : Z s) (hoff : ~ OnLine s) :
  ∃ u v, Z u ∧ Z v ∧ (psi u).rate < 0 ∧ 0 < (psi v).rate := by
  obtain (h, t) := s
  have hh : h ≠ 1 := hoff
  have hf : Z (2 - h, t) := hZ (h, t) hs
  by_cases hlt : h < 1
  · refine ((2 - h, t), (h, t), hf, hs, ?, ?)
    · show 1 - (2 - h) < 0; omega
    · show 0 < 1 - h; omega
  · refine ((h, t), (2 - h, t), hs, hf, ?, ?)

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  · show 1 - h < 0; omega
  · show 0 < 1 - (2 - h); omega

/-! ## Part VI. The posit has gravity's shape -/

/-- THE VALUE IS STATIONARITY. -/
theorem value_iff_stationary (Z : Pt → Prop) : Value Z ↔ ∀ s, Z s →
Stationary (psi s) :=
(fun h s hs => (line_is_stationary s).mpr (h s hs), fun h s hs =>
(line_is_stationary s).mpr (h s hs))

/-- Least erasure is stationarity. -/
theorem least_erasure_is_stationary (Z : Pt → Prop) :
LeastErasure Z ↔ ∀ s, Z s → Stationary (psi s) :=
(least_erasure_iff_value Z).trans (value_iff_stationary Z)

/-- The posit's shape is the fold: least erasure holds exactly when
the fold fixes every member,
  no member standing off the line and no remanent side, as gravity
  registers. -/
theorem posit_is_the_fold (Z : Pt → Prop) : LeastErasure Z ↔ ∀ s, Z
s → fold s = s :=
(least_erasure_iff_value Z).trans
(fun h s hs => (seat_fixed_line s).mpr (h s hs), fun h s hs =>
(seat_fixed_line s).mpr (h s hs))

/-- THE ACT IS THE VALUE: the premise that every zero is a
stationary mode is the value itself. -/
theorem act_is_the_value (Z : Pt → Prop) :
(∀ s, Z s → Stationary (psi s)) ↔ Value Z :=
(value_iff_stationary Z).symm

def ActuatedByEnergies (Z : Pt → Prop) : Prop := ∀ s, Z s → ∃ a :
Int, phi (a, 0) = s

/-- The crossing from the physical side, exact both ways. -/
theorem value_iff_actuated (Z : Pt → Prop) : Value Z ↔
ActuatedByEnergies Z := by
  constructor
  · intro h s hs
    have hl : s.1 = 1 := h s hs
    refine (s.2, ?)
    obtain (u, v) := s
    show ((1 - 0, v) : Pt) = (u, v)
    rw [pe]; exact (by simp at hl; omega, rfl)
  · intro h s hs
    obtain (a, e) := h s hs
    exact e ▸ measured_on_line a

/-- What a sentence over a larger class adds to a crossing is
exactly the unactuated. -/
theorem surplus_is_the_unactuated (Zact Zall : Pt → Prop) (hv :
Value Zact)
  (he : ∃ s, Zall s ∧ ~ OnLine s) : ∃ s, Zall s ∧ ~ Zact s ∧ ~
OnLine s := by
  obtain (s, hs, hoff) := he
  exact (s, hs, fun ha => hoff (hv s ha), hoff)

/-! ## Part VII. The hardware closes -/

structure Weyl where
  name : String
  mult : Int
  y6 : Int
  triplet : Bool
  doublet : Bool

/-- One generation as left-handed Weyl fields: Q, uc, dc, L, ec;
hypercharge times six. -/
def generation : List Weyl :=
[("Q", 6, 1, true, true), ("uc", 3, -4, true, false), ("dc", 3,
2, true, false),
("L", 2, -3, false, true), ("ec", 1, 6, false, false)]

def tsum (f : Weyl → Int) : List Weyl → Int
| [] => 0
| w :: ws => f w + tsum f ws

theorem tsum_append (f : Weyl → Int) (l1 l2 : List Weyl) :
tsum f (l1 ++ l2) = tsum f l1 + tsum f l2 := by
  induction l1 with
  | nil => simp [tsum]
  | cons w ws ih => simp [tsum, ih, Int.add_assoc]

def gravAnom (w : Weyl) : Int := w.mult * w.y6
def cubeAnom (w : Weyl) : Int := w.mult * w.y6 ^ 3
def colourAnom (w : Weyl) : Int := if w.triplet then (w.mult / 3) *
w.y6 else 0
def isospinAnom (w : Weyl) : Int := if w.doublet then (w.mult / 2) *
w.y6 else 0
def doubletCount (w : Weyl) : Int := if w.doublet then w.mult / 2
else 0

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def colourCube (w : Weyl) : Int :=
  if w.triplet then (if w.name = "Q" then w.mult / 3 else -(w.mult / 3)) else 0

/-- One generation closes: every gauge anomaly and the mixed
gravitational anomaly vanish, and
the isospin doublets number four, an even count. -/
theorem generation_closes :
  tsum gravAnom generation = 0 ∧ tsum cubeAnom generation = 0 ∧
  tsum colourAnom generation = 0 ∧ tsum isospinAnom generation = 0
  ∧
  tsum doubletCount generation = 4 ∧ tsum colourCube generation = 0
  := by decide

def gens : Nat → List Weyl
| 0 => []
| n + 1 => generation ++ gens n

theorem gens_sum (f : Weyl → Int) (n : Nat) : tsum f (gens n) = n *
tsum f generation := by
  induction n with
  | zero => simp [gens, tsum]
  | succ n ih => rw [gens, tsum_append, ih, Int.natCast_succ,
    Int.add_mul, Int.one_mul, Int.add_comm]

/-- Every number of generations closes: the anomaly conditions do
not fix the count. -/
theorem every_generation_count_closes (n : Nat) :
  tsum gravAnom (gens n) = 0 ∧ tsum cubeAnom (gens n) = 0 ∧
  tsum colourAnom (gens n) = 0 ∧ tsum isospinAnom (gens n) = 0 ∧
  tsum doubletCount (gens n) = 4 * n ∧ tsum colourCube (gens n) = 0
  := by
  obtain (g1, g2, g3, g4, g5, g6) := generation_closes
  refine {_, ?, _, ?, ?, ?} <:=> rw [gens_sum]
  · rw [g1, Int.mul_zero]
  · rw [g2, Int.mul_zero]
  · rw [g3, Int.mul_zero]
  · rw [g4, Int.mul_zero]
  · rw [g5, Int.mul_comm]
  · rw [g6, Int.mul_zero]

/-- The hypercharges are forced, not fitted. -/
theorem hypercharge_forced (yQ yu yd yL yE yH : Int)
  (Yu : yQ + yu + yH = 0) (Yd : yQ + yd - yH = 0) (Ye : yL + yE -
  yH = 0)
  (iso : 3 * yQ + yL = 0) (grav : 6 * yQ + 3 * yu + 3 * yd + 2 *
  yL + yE = 0) :
  yu = -4 * yQ ∧ yd = 2 * yQ ∧ yL = -3 * yQ ∧ yE = 6 * yQ ∧ yH = 3
  * yQ :=
  (by omega, by omega, by omega, by omega, by omega)

/-- Electric charge Q = T3 + Y, times six. -/
def charge6 (t3x6 y6 : Int) : Int := t3x6 + y6
theorem charges :
  charge6 3 1 = 4 ∧ charge6 (-3) 1 = -2 ∧ charge6 3 (-3) = 0 ∧
  charge6 (-3) (-3) = -6 := by decide
theorem matter_neutral :
  charge6 3 1 + charge6 3 1 + charge6 (-3) 1 = 6 ∧
  charge6 3 1 + charge6 (-3) 1 + charge6 (-3) 1 = 0 ∧
  (charge6 3 1 + charge6 3 1 + charge6 (-3) 1) + charge6 (-3) (-3)
  = 0 := by decide
theorem vacuum_neutral : charge6 (-3) 3 = 0 := by decide

/-- The colour lock: three colour labels carry a nonzero alternating
product exactly when they
are distinct, so a colour singlet of three quarks exists only on
three distinct axes. -/
def lockProduct (a b c : Fin 3) : Int :=
  (((b : Nat) : Int) - ((a : Nat) : Int)) * (((c : Nat) : Int) - ((a
  : Nat) : Int)) *
  (((c : Nat) : Int) - ((b : Nat) : Int))

theorem lock_iff_distinct :
  ∀ a b c : Fin 3, lockProduct a b c ≠ 0 ↔ (a ≠ b ∧ a ≠ c ∧ b ≠ c)
  := by decide

theorem lock_fails_when_colours_coincide : ∀ a c : Fin 3,
lockProduct a a c = 0 := by decide

/-! ## Part VIII. The dark sector, the neutrino's bit and floor,
least erasure as reversibility -/

/-- The dark sector: a stable particle fixed by the particle map
lands on the line under the
energy carrier, and under the charge carrier too, with every
charge reading zero. -/
theorem dark_on_the_line (c : Particle) (hfix : bar c = c) (hst :
IsReal c.energy) :
  OnLine (energySeat c) ∧ OnLine (chargeSeat c) ∧
  c.q3 = 0 ∧ c.colour = 0 ∧ c.b3 = 0 ∧ c.lepton = 0 := by
  obtain (h1, h2, h3, h4) := (bar_fixed_iff_neutral c).mp hfix
  exact (stable_lands c hst, neutral_charge_on_line c h1, h2,
  h3, h4)

def nuDirac (E : Energy) : Particle := (0, 0, 0, 1, E)
def nuMajorana (E : Energy) : Particle := (0, 0, 0, 0, E)

/-- The neutrino's one keyed bit: one gauge record, one seat point,
fixed in one world only. -/
theorem majorana_is_one_keyed_bit (E : Energy) :
  ((nuDirac E).q3 = (nuMajorana E).q3 ∧ (nuDirac E).colour =
  (nuMajorana E).colour ∧
  energySeat (nuDirac E) = energySeat (nuMajorana E) ∧
  chargeSeat (nuDirac E) = chargeSeat (nuMajorana E)) ∧
  bar (nuMajorana E) = nuMajorana E ∧ bar (nuDirac E) ≠ nuDirac E
  := by
  refine {(rfl, rfl, rfl, rfl), rfl, fun h => ?_}
  have h4 := congrArg Particle.lepton h
  change (-1 : Int) = 1 at h4
  omega

/-- The floor supplied by deed: a positive difference of squared
masses forbids a vanishing mass. -/
theorem floor_by_deed (m2 m3 : Int) (h : m2 * m2 < m3 * m3) : m3 ≠ 0
:= by
  intro e; subst e; have := sq_nonneg m2; rw [Int.mul_zero] at h;
  omega

def Merges {α : Type} (f : α → α) : Prop := ∃ x y, x ≠ y ∧ f x = f y

theorem reversible_merges_nothing {α : Type} (f g : α → α) (h : ∀ x,
g (f x) = x) :
  ¬ Merges f := by
  intro (x, y, hne, e)
  apply hne
  rw [← h x, ← h y, e]

theorem merging_is_irreversible {α : Type} (f : α → α) (hm : Merges
f) :
  ¬ ∃ g : α → α, ∀ x, g (f x) = x :=
  fun (g, h) => reversible_merges_nothing f g h hm

theorem mirror_is_reversible : ¬ Merges Energy.mirror :=
reversible_merges_nothing Energy.mirror Energy.mirror
mirror_involutive

/-- Registration merges exactly the off-line pair: it erases the
side of a point off the line. -/
theorem registration_erases_off_line (p : Pt) (hp : ¬ OnLine p) :
Merges reg :=
  (p, fold p, fun e => hp ((seat_fixed_line p).mp e.symm),
  (reg_forgets_side p).symm)

/-! ## Part IX. The capstones -/

/-- THE SUBSTRATE CARRIES THE SEAT: the fold is a global seat; both
carriers are equivariant, the
energy carrier for the time mirror and the charge carrier for
the particle map; the two
involutions commute; the energy carrier is blind to the particle
map; and a particle fixed by
both lands on the line under both. -/
theorem substrate_carries_the_seat :
  GlobalSeat OnLine fold ∧
  Equivariant phi Energy.mirror fold ∧ Equivariant chargeSeat bar
  fold ∧
  Equivariant energySeat tmirror fold ∧
  (∀ c : Particle, bar (tmirror c) = tmirror (bar c)) ∧
  (∀ c : Particle, energySeat (bar c) = energySeat c) ∧
  (∀ c : Particle, bar c = c → tmirror c = c → OnLine (energySeat
  c) ∧ OnLine (chargeSeat c)) :=
  (fold_global_seat, phi_equivariant, charge_carrier_equivariant,
  tmirror_equivariant,
  involutions_commute, pair_lands_once, fun c h1 h2 =>
  let hj := (joint_fixed_iff c).mp (h1, h2)
  (stable_lands c hj.2, neutral_charge_on_line c hj.1.1))

/-- THE PHYSICAL WITNESS OF THE CLOSURE. -/
theorem physical_witness :
  (∀ E : Energy, phi E.mirror = fold (phi E)) ∧
  (∀ E : Energy, psi (phi E) = E) ∧ (∀ p : Pt, phi (psi p) = p) ∧
  (∀ p : Pt, OnLine p ↔ Stationary (psi p)) ∧
  (∀ Z : Pt → Prop, Value Z ↔ ∀ s, Z s → Stationary (psi s)) ∧
  (∀ Z : Pt → Prop, Value Z ↔ ActuatedByEnergies Z) ∧
  (∀ Z : Pt → Prop, LeastErasure Z ↔ ∀ s, Z s → fold s = s) ∧
  (∃ Z : Pt → Prop, (∀ s, Z s → Z (fold s)) ∧
  (psi (0, 5)).rate + (psi (2, 5)).rate = 0 ∧ Z (0, 5) ∧ Z (2,
  5) ∧ ¬ Value Z) ∧
  (∀ F : (Pt → Prop) → Prop,
  (F (recordOf worldLine) ↔ F (recordOf worldPair)) ∧ Value
  worldLine ∧ ¬ Value worldPair) ∧
  (∀ H : Block, 0 ≤ H.disc ∧ H.T.disc = H.disc) :=
  (phi_equivariant, psi_phi, phi_psi, line_is_stationary,
  value_iff_stationary,

```

```

value_iff_actuated, posit_is_the_fold,
whole_conservation_forces_nothing,
record_decides_nothing, stationarity_survives_the_magnet)

```

```

/-- THE HARDWARE CLOSES. -/
theorem hardware_closes :

```

```

(tsum gravAnom generation = 0  $\wedge$  tsum cubeAnom generation = 0  $\wedge$ 
 tsum colourAnom generation = 0  $\wedge$  tsum isospinAnom generation =
 0  $\wedge$ 
 tsum doubletCount generation = 4  $\wedge$  tsum colourCube generation =
 0)  $\wedge$ 
( $\forall$  n : Nat, tsum gravAnom (gens n) = 0  $\wedge$  tsum cubeAnom (gens n) =
 0  $\wedge$ 
 tsum colourAnom (gens n) = 0  $\wedge$  tsum isospinAnom (gens n) = 0  $\wedge$ 
 tsum doubletCount (gens n) = 4 * n  $\wedge$  tsum colourCube (gens n) =
 0)  $\wedge$ 
(charge6 3 1 + charge6 3 1 + charge6 (-3) 1) + charge6 (-3) (-3)
= 0  $\wedge$ 
charge6 (-3) 3 = 0  $\wedge$ 
( $\forall$  a b c : Fin 3, lockProduct a b c  $\neq$  0  $\leftrightarrow$  (a  $\neq$  b  $\wedge$  a  $\neq$  c  $\wedge$  b  $\neq$ 
c)) :=
(generation_closes, every_generation_count_closes,
matter_neutral.2.2, vacuum_neutral,
lock_iff_distinct)

```

```

end SPHYS.OneSubstrate

```

```

#print axioms SPHYS.OneSubstrate.pe
#print axioms SPHYS.OneSubstrate.fold_involutive
#print axioms SPHYS.OneSubstrate.seat_fixed_line
#print axioms SPHYS.OneSubstrate.reg_lands
#print axioms SPHYS.OneSubstrate.reg_fixes_iff
#print axioms SPHYS.OneSubstrate.reg_forgets_side
#print axioms SPHYS.OneSubstrate.least_erasure_iff_value
#print axioms SPHYS.OneSubstrate.equivariant_id
#print axioms SPHYS.OneSubstrate.equivariant_comp
#print axioms SPHYS.OneSubstrate.fix_functorial
#print axioms SPHYS.OneSubstrate.fold_global_seat
#print axioms SPHYS.OneSubstrate.equivariant_carrier_lands
#print axioms SPHYS.OneSubstrate.value_on_image
#print axioms SPHYS.OneSubstrate.kinetic_crossing
#print axioms SPHYS.OneSubstrate.off_locus_pair
#print axioms SPHYS.OneSubstrate.ee
#print axioms SPHYS.OneSubstrate.mirror_involutive
#print axioms SPHYS.OneSubstrate.psi_phi
#print axioms SPHYS.OneSubstrate.phi_psi
#print axioms SPHYS.OneSubstrate.phi_injective
#print axioms SPHYS.OneSubstrate.phi_equivariant
#print axioms SPHYS.OneSubstrate.mirror_fixed_iff_real
#print axioms SPHYS.OneSubstrate.phi_line_iff_real
#print axioms SPHYS.OneSubstrate.stationary_iff_real
#print axioms SPHYS.OneSubstrate.line_is_stationary
#print axioms SPHYS.OneSubstrate.measured_on_line
#print axioms SPHYS.OneSubstrate.no_measurement_off_line

```

```

'SPHYS.OneSubstrate.pe' does not depend on any axioms
'SPHYS.OneSubstrate.fold_involutive' depends on axioms: [propxt,
Quot.sound]
'SPHYS.OneSubstrate.seat_fixed_line' depends on axioms: [propxt,
Quot.sound]
'SPHYS.OneSubstrate.reg_lands' does not depend on any axioms
'SPHYS.OneSubstrate.reg_fixes_iff' depends on axioms: [propxt]
'SPHYS.OneSubstrate.reg_forgets_side' does not depend on any axioms
'SPHYS.OneSubstrate.least_erasure_iff_value' depends on axioms:
[propxt]
'SPHYS.OneSubstrate.equivariant_id' does not depend on any axioms
'SPHYS.OneSubstrate.equivariant_comp' does not depend on any axioms
'SPHYS.OneSubstrate.fix_functorial' does not depend on any axioms
'SPHYS.OneSubstrate.fold_global_seat' depends on axioms: [propxt,
Quot.sound]
'SPHYS.OneSubstrate.equivariant_carrier_lands' depends on axioms:
[propxt, Quot.sound]
'SPHYS.OneSubstrate.value_on_image' does not depend on any axioms
'SPHYS.OneSubstrate.kinetic_crossing' does not depend on any axioms
'SPHYS.OneSubstrate.off_locus_pair' depends on axioms: [propxt,
Quot.sound]
'SPHYS.OneSubstrate.ee' does not depend on any axioms
'SPHYS.OneSubstrate.mirror_involutive' depends on axioms: [propxt,
Quot.sound]
'SPHYS.OneSubstrate.psi_phi' depends on axioms: [propxt, Quot.sound]
'SPHYS.OneSubstrate.phi_psi' depends on axioms: [propxt, Quot.sound]
'SPHYS.OneSubstrate.phi_injective' depends on axioms: [propxt,
Quot.sound]
'SPHYS.OneSubstrate.phi_equivariant' depends on axioms: [propxt,
Quot.sound]
'SPHYS.OneSubstrate.mirror_fixed_iff_real' depends on axioms:
[propxt, Quot.sound]
'SPHYS.OneSubstrate.phi_line_iff_real' depends on axioms: [propxt,
Quot.sound]
'SPHYS.OneSubstrate.stationary_iff_real' depends on axioms: [propxt]
'SPHYS.OneSubstrate.line_is_stationary' depends on axioms: [propxt,
Quot.sound]

```

```

#print axioms SPHYS.OneSubstrate.energy_carrier_lands
#print axioms SPHYS.OneSubstrate.bar_involutive
#print axioms SPHYS.OneSubstrate.bar_fixed_iff_neutral
#print axioms SPHYS.OneSubstrate.odd_charges_even_energy
#print axioms SPHYS.OneSubstrate.gravity_reads_no_charge_bit
#print axioms SPHYS.OneSubstrate.involutions_commute
#print axioms SPHYS.OneSubstrate.tmirror_involutive
#print axioms SPHYS.OneSubstrate.joint_fixed_iff
#print axioms SPHYS.OneSubstrate.charge_carrier_equivariant
#print axioms SPHYS.OneSubstrate.pair_lands_once
#print axioms SPHYS.OneSubstrate.tmirror_equivariant
#print axioms SPHYS.OneSubstrate.neutral_charge_on_line
#print axioms SPHYS.OneSubstrate.pair_on_the_edges
#print axioms SPHYS.OneSubstrate.stable_lands
#print axioms SPHYS.OneSubstrate.off_line_pair
#print axioms SPHYS.OneSubstrate.resonance_pair
#print axioms SPHYS.OneSubstrate.sq_nonneg
#print axioms SPHYS.OneSubstrate.block_T_fixed_iff
#print axioms SPHYS.OneSubstrate.stationarity_survives_the_magnet
#print axioms SPHYS.OneSubstrate.worlds_share_the_record
#print axioms SPHYS.OneSubstrate.world_line_value
#print axioms SPHYS.OneSubstrate.world_pair_not_value
#print axioms SPHYS.OneSubstrate.record_decides_nothing
#print axioms SPHYS.OneSubstrate.pair_world_fold_closed
#print axioms SPHYS.OneSubstrate.pair_world_total_rate
#print axioms SPHYS.OneSubstrate.whole_conservation_forces_nothing
#print axioms
SPHYS.OneSubstrate.fold_closed_off_line_decays_and_grows
#print axioms SPHYS.OneSubstrate.value_iff_stationary
#print axioms SPHYS.OneSubstrate.least_erasure_is_stationarity
#print axioms SPHYS.OneSubstrate.posit_is_the_fold
#print axioms SPHYS.OneSubstrate.act_is_the_value
#print axioms SPHYS.OneSubstrate.value_iff_actuated
#print axioms SPHYS.OneSubstrate.surplus_is_the_unactuated
#print axioms SPHYS.OneSubstrate.tsum_append
#print axioms SPHYS.OneSubstrate.generation_closes
#print axioms SPHYS.OneSubstrate.gens_sum
#print axioms SPHYS.OneSubstrate.every_generation_count_closes
#print axioms SPHYS.OneSubstrate.hypercharge_forced
#print axioms SPHYS.OneSubstrate.charges
#print axioms SPHYS.OneSubstrate.matter_neutral
#print axioms SPHYS.OneSubstrate.vacuum_neutral
#print axioms SPHYS.OneSubstrate.lock_iff_distinct
#print axioms SPHYS.OneSubstrate.lock_fails_when_colours_coincide
#print axioms SPHYS.OneSubstrate.dark_on_the_line
#print axioms SPHYS.OneSubstrate.majorana_is_one_keyed_bit
#print axioms SPHYS.OneSubstrate.floor_by_deed
#print axioms SPHYS.OneSubstrate.reversible_merges_nothing
#print axioms SPHYS.OneSubstrate.merging_is_irreversible
#print axioms SPHYS.OneSubstrate.mirror_is_reversible
#print axioms SPHYS.OneSubstrate.registration_erases_off_line
#print axioms SPHYS.OneSubstrate.substrate_carries_the_seat
#print axioms SPHYS.OneSubstrate.physical_witness
#print axioms SPHYS.OneSubstrate.hardware_closes

```

```

'SPHYS.OneSubstrate.measured_on_line' does not depend on any axioms
'SPHYS.OneSubstrate.no_measurement_off_line' does not depend on any
axioms
'SPHYS.OneSubstrate.energy_carrier_lands' depends on axioms:
[propxt, Quot.sound]
'SPHYS.OneSubstrate.bar_involutive' does not depend on any axioms
'SPHYS.OneSubstrate.bar_fixed_iff_neutral' depends on axioms:
[propxt, Quot.sound]
'SPHYS.OneSubstrate.odd_charges_even_energy' does not depend on any
axioms
'SPHYS.OneSubstrate.gravity_reads_no_charge_bit' does not depend on
any axioms
'SPHYS.OneSubstrate.involutions_commute' does not depend on any
axioms
'SPHYS.OneSubstrate.tmirror_involutive' depends on axioms: [propxt,
Quot.sound]
'SPHYS.OneSubstrate.joint_fixed_iff' depends on axioms: [propxt,
Quot.sound]
'SPHYS.OneSubstrate.charge_carrier_equivariant' depends on axioms:
[propxt, Quot.sound]
'SPHYS.OneSubstrate.pair_lands_once' does not depend on any axioms
'SPHYS.OneSubstrate.tmirror_equivariant' depends on axioms:
[propxt, Quot.sound]
'SPHYS.OneSubstrate.neutral_charge_on_line' depends on axioms:
[propxt, Quot.sound]
'SPHYS.OneSubstrate.pair_on_the_edges' does not depend on any axioms
'SPHYS.OneSubstrate.stable_lands' depends on axioms: [propxt,
Quot.sound]
'SPHYS.OneSubstrate.off_line_pair' depends on axioms: [propxt,
Quot.sound]
'SPHYS.OneSubstrate.resonance_pair' depends on axioms: [propxt,
Quot.sound]
'SPHYS.OneSubstrate.sq_nonneg' depends on axioms: [propxt,
Quot.sound]
'SPHYS.OneSubstrate.block_T_fixed_iff' depends on axioms: [propxt,
Quot.sound]

```

```

'SPHYS.OneSubstrate.stationarity_survives_the_magnet' depends on
axioms: [propext, Quot.sound]
'SPHYS.OneSubstrate.worlds_share_the_record' does not depend on any
axioms
'SPHYS.OneSubstrate.world_line_value' does not depend on any axioms
'SPHYS.OneSubstrate.world_pair_not_value' does not depend on any
axioms
'SPHYS.OneSubstrate.record_decides_nothing' depends on axioms:
[propext, Quot.sound]
'SPHYS.OneSubstrate.pair_world_fold_closed' does not depend on any
axioms
'SPHYS.OneSubstrate.pair_world_total_rate' does not depend on any
axioms
'SPHYS.OneSubstrate.whole_conservation_forces_nothing' does not
depend on any axioms
'SPHYS.OneSubstrate.fold_closed_off_line_decays_and_grows' depends
on axioms: [propext, Quot.sound]
'SPHYS.OneSubstrate.value_iff_stationary' depends on axioms:
[propext, Quot.sound]
'SPHYS.OneSubstrate.least_erasure_is_stationarity' depends on
axioms: [propext, Quot.sound]
'SPHYS.OneSubstrate.posit_is_the_fold' depends on axioms: [propext,
Quot.sound]
'SPHYS.OneSubstrate.act_is_the_value' depends on axioms: [propext,
Quot.sound]
'SPHYS.OneSubstrate.value_iff_actuated' depends on axioms: [propext,
Quot.sound]
'SPHYS.OneSubstrate.surplus_is_the_unactuated' does not depend on
any axioms
'SPHYS.OneSubstrate.tsum_append' depends on axioms: [propext]
'SPHYS.OneSubstrate.generation_closes' does not depend on any axioms

```

```

'SPHYS.OneSubstrate.gens_sum' depends on axioms: [propext]
'SPHYS.OneSubstrate.every_generation_count_closes' depends on
axioms: [propext]
'SPHYS.OneSubstrate.hypercharge_forced' depends on axioms: [propext,
Quot.sound]
'SPHYS.OneSubstrate.charges' does not depend on any axioms
'SPHYS.OneSubstrate.matter_neutral' does not depend on any axioms
'SPHYS.OneSubstrate.vacuum_neutral' does not depend on any axioms
'SPHYS.OneSubstrate.lock_iff_distinct' does not depend on any axioms
'SPHYS.OneSubstrate.lock_fails_when_colours_coincide' does not
depend on any axioms
'SPHYS.OneSubstrate.dark_on_the_line' depends on axioms: [propext,
Quot.sound]
'SPHYS.OneSubstrate.majorana_is_one_keyed_bit' depends on axioms:
[propext, Quot.sound]
'SPHYS.OneSubstrate.floor_by_deed' depends on axioms: [propext,
Quot.sound]
'SPHYS.OneSubstrate.reversible_merges_nothing' does not depend on
any axioms
'SPHYS.OneSubstrate.merging_is_irreversible' does not depend on any
axioms
'SPHYS.OneSubstrate.mirror_is_reversible' depends on axioms:
[propext, Quot.sound]
'SPHYS.OneSubstrate.registration_erases_off_line' depends on axioms:
[propext, Quot.sound]
'SPHYS.OneSubstrate.substrate_carries_the_seat' depends on axioms:
[propext, Quot.sound]
'SPHYS.OneSubstrate.physical_witness' depends on axioms: [propext,
Quot.sound]
'SPHYS.OneSubstrate.hardware_closes' depends on axioms: [propext]

```

APPENDIX B: SUBSTRATE_TWIN.F90 AND ITS RUN

The twin, verbatim, and its run under gfortran -std=f2018 -O2 -fno-fast-math -ffp-contract=off.

```

! Substrate_Twin.f90 · the executed twin of SPHYS_One_Substrate.lean
· Fortran 2018.
! Build (sealed): gfortran -std=f2018 -O2 -fno-fast-math
-ffp-contract=off Substrate_Twin.f90
! Blocks: A the integer chart; B the exact complex carrier; C closed
systems and the magnet;
! D the balanced dimer; E the actuated zeros; F the hardware and the
colour lock; G the particle
! map, the two carriers, antimatter's fall and measured widths; H
the neutrino's floor; I the price.
program substrate_twin
  implicit none
  integer, parameter :: dp = selected_real_kind(15, 307)
  integer, parameter :: i8 = selected_int_kind(18)
  real(dp), parameter :: HBAR_EVS = 6.582119569e-16_dp, YEAR_S =
3.15576e7_dp
  real(dp), parameter :: KB = 1.380649e-23_dp
  integer :: checks, fails
  real(dp) :: pi
  checks = 0; fails = 0; pi = acos(-1.0_dp)

  call block_a(); call block_b(); call block_c(); call block_d()
  call block_e(); call block_f(); call block_g(); call block_h();
  call block_i()

  write(*, '(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', checks,
', "failures":', fails, '}'
  if (fails > 0) error stop 1
contains

  subroutine check(name, ok)
    character(*), intent(in) :: name
    logical, intent(in) :: ok
    checks = checks + 1
    if (.not. ok) fails = fails + 1
    write(*, '(a,a)') merge(' PASS ', ' FAIL ', ok), name
  end subroutine check

! -----
A. the integer chart
subroutine block_a()
  integer :: a, b, t, h, n, neq, nrt, nline, nstat, npair
  logical :: stat
  n = 0; neq = 0; nrt = 0; nline = 0; nstat = 0; npair = 0
  write(*, '(a)') 'A · the integer chart: E = (a, beta), phi(E) =
(1 - beta, a), fold (h, t) = (2 - h, t)'
  do a = -30, 30
    do b = -30, 30
      n = n + 1
      ! equivariance: phi(mirror E) = fold(phi E)
      if ((1 - (-b)) == 2 - (1 - b)) neq = neq + 1
      ! round trip: psi(phi E) = E, psi(h, t) = (t, 1 - h)

```

```

h = 1 - b
if (a == a .and. (1 - h) == b) nrt = nrt + 1
! the line is the image of the real energies
if ((h == 1) .eqv. (b == 0)) nline = nline + 1
! stationary: beta t = 0 at every integer time in [-12, 12]
exactly when beta = 0
stat = .true.
do t = -12, 12
  if (b*t /= 0) stat = .false.
end do
if (stat .eqv. (b == 0)) nstat = nstat + 1
! the off-line pair: one record, rates cancel, exactly one
decays
if (b /= 0) then
  if ((2 - h) /= h .and. b + (-b) == 0 .and. (b < 0) .neqv.
-b < 0)) npair = npair + 1
end if
end do
call check('equivariance on 3721 energies: the time mirror of
the rate is the fold', neq == n)
call check('the carrier is a bijection: psi(phi E) = E on 3721
energies', nrt == n)
call check('the line is exactly the image of the real energies
(3721 of 3721)', nline == n)
call check('stationary at every time in [-12,12] exactly when
real (3721 of 3721)', nstat == n)
call check('every off-line mode has a distinct mirror, one
record, rates cancelling, one decaying (3660)', &
npair == n - 61)
end subroutine block_a

! -----
B. the exact complex carrier
subroutine block_b()
  integer :: k, t
  integer(i8) :: seed
  complex(dp) :: E, s, sf, sm
  real(dp) :: dmax, x, y, nrm, dn_real, dn_off
  seed = 20260930_i8; dmax = 0.0_dp; dn_real = 0.0_dp; dn_off =
huge(1.0_dp)
  write(*, '(a)') 'B · the complex carrier s = 1/2 + iE: fold(s) = 1
- conj(s), mirror(E) = conj(E)'
  do k = 1, 20000
    x = 200.0_dp*(rnd(seed) - 0.5_dp); y = 2.0_dp*(rnd(seed) -
0.5_dp)
    E = cmplx(x, y, dp)
    s = cmplx(0.5_dp, 0.0_dp, dp) + cmplx(0.0_dp, 1.0_dp, dp)*E
    sf = cmplx(1.0_dp, 0.0_dp, dp) - conjg(s)
    sm = cmplx(0.5_dp, 0.0_dp, dp) + cmplx(0.0_dp, 1.0_dp,
dp)*conjg(E)
    dmax = max(dmax, abs(sf - sm))

```

```

end do
write(*,'(a,es10.3)') ' max |fold(s(E)) - s(conj E)| over 20000
energies: ', dmax
call check('fold(s(E)) = s(conj E) on 20000 complex energies, to
4 ulp of the scale', &
dmax <= 4.0_dp*epsilon(1.0_dp)*100.0_dp)
! the norm of e^{-iEt}: stationary exactly on the real axis
do k = 1, 2000
x = 50.0_dp*(rnd(seed) - 0.5_dp)
do t = 1, 10
nrm = abs(exp(-cmplx(0.0_dp, 1.0_dp, dp)*cmplx(x, 0.0_dp,
dp)*real(t, dp)))*2
dn_real = max(dn_real, abs(nrm - 1.0_dp))
nrm = abs(exp(-cmplx(0.0_dp, 1.0_dp, dp)*cmplx(x, 1.0e-3_dp,
dp)*real(t, dp)))*2
dn_off = min(dn_off, abs(nrm - 1.0_dp))
end do
end do
write(*,'(a,es10.3,a,es10.3)') ' real energies: max |norm - 1| =
', dn_real, &
'; rate 2e-3: min |norm - 1| = ', dn_off
call check('a real energy keeps its norm at every time to
1e-14', dn_real <= 1.0e-14_dp)
call check('a mode off the axis moves its norm (growth e^{beta
t}) visible at t >= 1', dn_off >= 1.0e-3_dp)
end subroutine block_b

! -----
C. closed systems
subroutine block_c()
integer :: k, nneg, nneg0, ntb
integer(i8) :: seed
real(dp) :: a, b, cr, ci, disc, disc0
seed = 777_i8; nneg = 0; nneg0 = 0; ntb = 0
write(*,'(a)') 'C · closed systems and the magnet: a Hermitian
generator has a real spectrum, T-broken or not'
do k = 1, 100000
a = 20.0_dp*(rnd(seed) - 0.5_dp); b = 20.0_dp*(rnd(seed) -
0.5_dp)
cr = 20.0_dp*(rnd(seed) - 0.5_dp); ci = 20.0_dp*(rnd(seed) -
0.5_dp)
disc = (a - b)**2 + 4.0_dp*(cr*cr + ci*ci)
disc0 = (a - b)**2 + 4.0_dp*(cr*cr)
if (ci /= 0.0_dp) ntb = ntb + 1
if (disc < 0.0_dp) nneg = nneg + 1
if (disc0 < 0.0_dp) nneg0 = nneg0 + 1
end do
write(*,'(a,i0,a)') ' time-broken blocks (imaginary
off-diagonal, the magnetic term): ', ntb, ' of 100000'
call check('100000 time-broken Hermitian blocks: discriminant >=
0, every eigenvalue real', nneg = 0 .and. ntb == 100000)
call check('their time-symmetric parts (off-diagonal made real):
every eigenvalue real', nneg0 == 0)
end subroutine block_c

! -----
D. the balanced dimer
subroutine block_d()
integer :: i, nreal, npair, nbal, ntot
real(dp) :: w, kap, gam, g
complex(dp) :: tr, det, sq, ll, l2
write(*,'(a)') 'D · the balanced gain-loss dimer H = [[w + i g,
k], [k, w - i g]]: open halves, balanced whole'
w = 1.0_dp; kap = 0.25_dp; nreal = 0; npair = 0; nbal = 0; ntot =
0
do i = 0, 200
gam = 0.005_dp*real(i, dp)
if (abs(gam - kap) < 1.0e-12_dp) cycle
ntot = ntot + 1
tr = cmplx(2.0_dp*w, 0.0_dp, dp)
det = cmplx(w, gam, dp)*cmplx(w, -gam, dp) - cmplx(kap*kap,
0.0_dp, dp)
sq = sqrt(tr*tr - 4.0_dp*det)
ll = 0.5_dp*(tr + sq); l2 = 0.5_dp*(tr - sq)
if (abs(aimag(ll) + aimag(l2)) <= 1.0e-14_dp) nbal = nbal + 1
if (gam < kap) then
if (abs(aimag(ll)) <= 1.0e-14_dp .and. abs(aimag(l2)) <=
1.0e-14_dp) nreal = nreal + 1
else
g = sqrt(gam*gam - kap*kap)
if (abs(real(ll, dp) - real(l2, dp)) <= 1.0e-14_dp .and.
abs(abs(aimag(ll)) - g) <= 1.0e-13_dp &
.and. aimag(ll)*aimag(l2) < 0.0_dp) npair = npair + 1
end if
end do
write(*,'(a,i0,a,i0,a,i0)') ' gain below coupling, real pairs:
', nreal, &
'; above, mirror pairs: ', npair, '; swept: ', ntot
call check('below the exceptional point both modes are real: on the
line (50 of 50)', nreal == 50)
call check('above it the modes form one mirror pair: equal
frequency, opposite rates (150 of 150)', npair == 150)

```

```

call check('the whole is balanced at every gain: total rate zero
(200 of 200)', nbal == ntot)
end subroutine block_d

! -----
E. the actuated zeros
pure function theta(t) result(r)
real(dp), intent(in) :: t
real(dp) :: r
r = 0.5_dp*t*log(t/(2.0_dp*acos(-1.0_dp))) - 0.5_dp*t -
acos(-1.0_dp)/8.0_dp &
+ 1.0_dp/(48.0_dp*t) + 7.0_dp/(5760.0_dp*t**3) +
31.0_dp/(80640.0_dp*t**5)
end function theta

! zeta(1/2 + it) by Euler-Maclaurin summation, N = 60 terms and
eight Bernoulli corrections;
! the truncation error at t <= 100 is below 1e-11. Hardy's Z(t) =
e^{i theta(t)} zeta(1/2 + it).
function zeta_em(t) result(z)
real(dp), intent(in) :: t
complex(dp) :: z, s, term, poch
integer, parameter :: NN = 60
real(dp), parameter :: B2K(8) = [1.0_dp/6.0_dp, -1.0_dp/30.0_dp,
1.0_dp/42.0_dp, -1.0_dp/30.0_dp, &
5.0_dp/66.0_dp, -691.0_dp/2730.0_dp, 7.0_dp/6.0_dp,
-3617.0_dp/510.0_dp]
real(dp) :: fact
integer :: n, k
s = cmplx(0.5_dp, t, dp)
z = (0.0_dp, 0.0_dp)
do n = 1, NN - 1
z = z + exp(-s*log(real(n, dp)))
end do
z = z + exp((1.0_dp - s)*log(real(NN, dp)))/(s - 1.0_dp) +
0.5_dp*exp(-s*log(real(NN, dp)))
poch = s; fact = 2.0_dp
do k = 1, 8
term = B2K(k)/fact*poch*exp(-(s + real(2*k - 1,
dp))*log(real(NN, dp)))
z = z + term
poch = poch*(s + real(2*k - 1, dp))*(s + real(2*k, dp))
fact = fact*real(2*k + 1, dp)*real(2*k + 2, dp)
end do
end function zeta_em

function zrs(t) result(r)
real(dp), intent(in) :: t
real(dp) :: r
r = real(exp(cmplx(0.0_dp, theta(t), dp))*zeta_em(t), dp)
end function zrs

function zimag(t) result(r)
real(dp), intent(in) :: t
real(dp) :: r
r = aimag(exp(cmplx(0.0_dp, theta(t), dp))*zeta_em(t))
end function zimag

subroutine block_e()
real(dp), parameter :: KNOWN(10) = [14.134725142_dp,
21.022039639_dp, 25.010857580_dp, &
30.424876126_dp, 32.935061588_dp, 37.586178159_dp,
40.918719012_dp, 43.327073281_dp, &
48.005150881_dp, 49.773832478_dp]
real(dp) :: t, dt, zl, zr, lo, hi, mid, found(40), emax,
nsmooth, imax
integer :: nz, k, it
write(*,'(a)') 'E · the actuated zeros: sign changes of Hardy's
Z on the line, heights 10 to 100'
dt = 0.01_dp; t = 10.0_dp; zl = zrs(t); nz = 0; imax = 0.0_dp
do while (t + dt <= 100.0_dp)
zr = zrs(t + dt)
imax = max(imax, abs(zimag(t + dt)))
if (zl*zr < 0.0_dp) then
nz = nz + 1
lo = t; hi = t + dt
do it = 1, 60
mid = 0.5_dp*(lo + hi)
if (zrs(lo)*zrs(mid) <= 0.0_dp) then
hi = mid
else
lo = mid
end if
end do
if (nz <= 40) found(nz) = 0.5_dp*(lo + hi)
end if
t = t + dt; zl = zr
end do
nsmooth = theta(100.0_dp)/acos(-1.0_dp) + 1.0_dp
emax = 0.0_dp
do k = 1, 10
emax = max(emax, abs(found(k) - KNOWN(k)))

```

```

end do
write(*,'(a,i0,a,f8.4)') ' sign changes on (10, 100]: ', nz, '
theta(100)/pi + 1 = ', nsmooth
write(*,'(a,f13.9,a,e9.2)') ' first zero at t = ', found(1),
'; max error on the first ten: ', emax
write(*,'(a,e9.2)') ' max |Im e^{i theta} zeta(1/2+it)| on the
scan (Z is real): ', imax
call check('Z(t) = e^{i theta} zeta(1/2 + it) is real on the
scan to 1e-8', imax <= 1.0e-8_dp)
call check('29 sign changes of Z on (10, 100]: 29 zeros located
on the line by the executed chart', nz == 29)
call check('the count agrees with the smooth term theta(100)/pi +
1, rounded (29)', nint(nsmooth) == 29)
call check('the first ten located heights match the tabulated
zeros to 1e-8', emax <= 1.0e-8_dp)
end subroutine block_e

! -----
F. the hardware
subroutine block_f()
integer, parameter :: MULT(5) = [6, 3, 3, 2, 1], Y6(5) = [1, -4,
2, -3, 6]
logical, parameter :: TRIP(5) = [.true., .true., .true.,
.false., .false.]
logical, parameter :: DOUB(5) = [.true., .false., .false.,
.true., .false.]
integer :: n, i, s1, s3, sc, sw, nd, cc, nok, yu, yd, yl, ye,
yh, nsol, sol(5)
write(*,'(a)') 'F · the hardware: generations, forced
hypercharges, charges'
nok = 0
do n = 1, 6
s1 = 0; s3 = 0; sc = 0; sw = 0; nd = 0; cc = 0
do i = 1, 5
s1 = s1 + n*MULT(i)*Y6(i)
s3 = s3 + n*MULT(i)*Y6(i)**3
if (TRIP(i)) sc = sc + n*(MULT(i)/3)*Y6(i)
if (DOUB(i)) sw = sw + n*(MULT(i)/2)*Y6(i)
if (DOUB(i)) nd = nd + n*(MULT(i)/2)
if (TRIP(i)) cc = cc + n*merge(MULT(i)/3, -(MULT(i)/3), i ==
1)
end do
if (s1 == 0 .and. s3 == 0 .and. sc == 0 .and. sw == 0 .and. nd
== 4*n .and. cc == 0) nok = nok + 1
end do
call check('one to six generations: every anomaly vanishes,
doublets 4n (the count is not forced)', nok == 6)
nsol = 0
do yu = -12, 12
do yd = -12, 12
do yl = -12, 12
do ye = -12, 12
do yh = -12, 12
if (1 + yu + yh == 0 .and. 1 + yd - yh == 0 .and. yl +
ye - yh == 0 .and. &
3 + yl == 0 .and. 6 + 3*yu + 3*yd + 2*yl + ye ==
0) then
nsol = nsol + 1; sol = [yu, yd, yl, ye, yh]
end if
end do
end do
end do
end do
end do
call check('at y0 = 1 the five conditions admit one hypercharge
vector in [-12,12]^5: (-4, 2, -3, 6, 3)', &
nsol == 1 .and. all(sol == [-4, 2, -3, 6, 3]))
call check('charges Q = T3 + Y: up 2/3, down -1/3, neutrino 0,
electron -1', &
3 + 1 == 4 .and. -3 + 1 == -2 .and. 3 - 3 == 0 .and.
-3 - 3 == -6)
call check('proton one unit, neutron none, hydrogen neutral,
vacuum neutral', &
4 + 4 - 2 == 6 .and. 4 - 2 - 2 == 0 .and. 4 + 4 - 2 -
6 == 0 .and. -3 + 3 == 0)
block
integer :: ca, cb, cc2, nlock, nok2
nlock = 0; nok2 = 0
do ca = 0, 2
do cb = 0, 2
do cc2 = 0, 2
if ((cb - ca)*(cc2 - ca)*(cc2 - cb) /= 0) nlock = nlock +
1
if (((cb - ca)*(cc2 - ca)*(cc2 - cb) /= 0) .eqv. (ca /=
cb .and. ca /= cc2 .and. cb /= cc2)) &
nok2 = nok2 + 1
end do
end do
end do
call check('the colour lock: of 27 colour triples exactly the 6
on three distinct axes are nonzero', &
nlock == 6 .and. nok2 == 27)

```

```

end block
end subroutine block_f

! -----
G. the particle map
subroutine block_g()
integer, parameter :: NP = 9
! q3, colour, b3, lepton, width flag (0 stable in the table)
integer, parameter :: TAB(4, NP) = reshape([ -3, 0, 0, 1, 3,
0, 3, 0, 0, 0, 3, 0, &
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,
0, 0, 0, -3, 0, 0, 1 ], [4, NP])
! electron, proton, neutron, photon, Z, neutrino (Dirac),
neutrino (Majorana), dark stable, muon
logical, parameter :: NEUTRAL_EXPECT(NP) = [.false., .false.,
.false., .true., .true., .false., &
.true., .true., .false.]
integer :: k, nfix, ninv
real(dp) :: ge, gp, gmu, gz, gn, me, mmu, mz, mn
write(*,'(a)') 'G · the particle map: every additive charge
flipped, the energy kept'
nfix = 0; ninv = 0
do k = 1, NP
if (all(-(TAB(:, k)) == TAB(:, k))) ninv = ninv + 1
if (all(-(TAB(:, k)) == TAB(:, k))) .eqv. NEUTRAL_EXPECT(k))
nfix = nfix + 1
end do
call check('the particle map is an involution on the nine-entry
table', ninv == NP)
call check('fixed exactly on the neutral entries: photon, Z,
Majorana neutrino, the dark state', nfix == NP)
call check('Dirac and Majorana neutrinos share the gauge record
and differ in the one bit', &
TAB(1, 6) == TAB(1, 7) .and. TAB(2, 6) == TAB(2, 7)
.and. TAB(4, 6) /= TAB(4, 7))
block
integer :: q, fr, bt, ncar, ncom, njoint, nlep
logical :: fixc, fixt
ncar = 0; ncom = 0; njoint = 0; nlep = 0
do q = -9, 9
do fr = -20, 20
! charge carrier (1 + q3, freq): the particle map goes to
the fold
if (1 + (-q) == 2 - (1 + q)) ncar = ncar + 1
do bt = -3, 3
! the two involutions commute on (q, fr, bt): bar flips
q, the mirror flips bt
! bar then mirror against mirror then bar, composed on
the tuple (q, fr, bt)
if (all(mirror_t(bar_t([q, fr, bt])) ==
bar_t(mirror_t([q, fr, bt]))) ncom = ncom + 1
fixc = (q == 0); fixt = (bt == 0)
if ((fixc .and. fixt) .eqv. ((1 + q == 1) .and. (1 - bt
== 1))) njoint = njoint + 1
end do
end do
if (1 + (-1) == 0 .and. 1 + 1 == 2 .and. 2 - 0 == 2) nlep = 1
call check('the charge carrier sends the particle map onto the
fold (779 of 779)', ncar == 19*41)
call check('the two involutions commute; fixed by both exactly
when on the line under both (5453)', &
ncom == 19*41*7 .and. njoint == 19*41*7)
call check('the electron and the positron land on the edges h =
0 and h = 2, mirror images', nlep == 1)
end block
block
real(dp) :: ag, sg
ag = 0.75_dp; sg = sqrt(0.13_dp**2 + 0.16_dp**2)
write(*,'(a,f5.2,a)') ' antihydrogen falls with (0.75 +- 0.13
+- 0.16) g: from g at ', (1.0_dp - ag)/sg, ' sigma'
call check('gravity reads no charge bit: antimatter falls as
matter within 2 sigma (ALPHA-g)', (1.0_dp - ag)/sg < 2.0_dp)
end block
! measured widths, hbar / tau, and their distance from the line
relative to the rest energy
me = 510998.95_dp; mmu = 105.6583755e6_dp; mz = 91.1876e9_dp; mn
= 939.56542052e6_dp
ge = HBAR_EV/ (6.6e28_dp*YEAR_S)
gp = HBAR_EV/ (2.4e34_dp*YEAR_S)
gmu = HBAR_EV/ 2.1969811e-6_dp
gn = HBAR_EV/ 878.4_dp
gz = 2.4952e9_dp
write(*,'(a,es10.3,a,es10.3)') ' electron (e -> gamma nu):
width < ', ge, ' eV; width/m < ', ge/me
write(*,'(a,es10.3,a,es10.3)') ' proton (p -> e+ pi0):
width < ', gp, ' eV; width/m < ', gp/938.27208816e6_dp
write(*,'(a,es10.3,a,es10.3)') ' neutron (free):
width = ', gn, ' eV; width/m = ', gn/mn
write(*,'(a,es10.3,a,es10.3)') ' muon:
width = ', gmu, ' eV; width/m = ', gmu/mmu
write(*,'(a,es10.3,a,es10.3)') ' Z boson:
width = ', gz, ' eV; width/m = ', gz/mz

```

```

call check('the electron stands on the line to 1e-57 of its rest
energy (partial width bound)', ge/me < 1.0e-57_dp)
call check('the proton stands on the line to 1e-65 of its rest
energy (partial width bound)', &
gp/938.27208816e6_dp < 1.0e-65_dp)
call check('the muon, neutron and Z stand off the line: open
modes whose decay exports to products', &
gmu > 0.0_dp .and. gn > 0.0_dp .and. gz/mz > 0.02_dp)
end subroutine block_g

! -----
H. the neutrino's floor
subroutine block_h()
real(dp) :: d21, d32, m2min, m3min
write(*,'(a)') 'H · the neutrino's floor, supplied by deed:
oscillation returns positive mass-squared differences'
d21 = 7.53e-5_dp; d32 = 2.455e-3_dp
m2min = sqrt(d21); m3min = sqrt(d21 + d32)
write(*,'(a,es10.3,a,es10.3,a)') ' lightest possible m2 = ',
m2min, ' eV; m3 >= ', m3min, ' eV (normal ordering)'
call check('both measured splittings are positive, so at least
two neutrino masses are nonzero', &
d21 > 0.0_dp .and. d32 > 0.0_dp .and. m2min >
8.0e-3_dp .and. m3min > 4.9e-2_dp)
end subroutine block_h

! -----
I. the price
subroutine block_i()
real(dp) :: q, qexact

A · the integer chart: E = (a, beta), phi(E) = (1 - beta, a), fold
(h, t) = (2 - h, t)
PASS equivariance on 3721 energies: the time mirror of the rate
is the fold
PASS the carrier is a bijection: psi(phi E) = E on 3721 energies
PASS the line is exactly the image of the real energies (3721 of
3721)
PASS stationary at every time in [-12,12] exactly when real (3721
of 3721)
PASS every off-line mode has a distinct mirror, one record, rates
cancelling, one decaying (3660)
B · the complex carrier s = 1/2 + iE: fold(s) = 1 - conj(s),
mirror(E) = conj(E)
max |fold(s(E)) - s(conj E)| over 20000 energies: 1.110E-16
PASS fold(s(E)) = s(conj E) on 20000 complex energies, to 4 ulp
of the scale
real energies: max |norm - 1| = 2.220E-16; rate 2e-3: min |norm -
1| = 2.002E-03
PASS a real energy keeps its norm at every time to 1e-14
PASS a mode off the axis moves its norm (growth e^{beta t})
visible at t >= 1)
C · closed systems and the magnet: a Hermitian generator has a real
spectrum, T-broken or not
time-broken blocks (imaginary off-diagonal, the magnetic term):
100000 of 100000
PASS 100000 time-broken Hermitian blocks: discriminant >= 0,
every eigenvalue real
PASS their time-symmetric parts (off-diagonal made real): every
eigenvalue real
D · the balanced gain-loss dimer H = [[w + i g, k], [k, w - i g]]:
open halves, balanced whole
gain below coupling, real pairs: 50; above, mirror pairs: 150;
swept: 200
PASS below the exceptional point both modes are real: on the line
(50 of 50)
PASS above it the modes form one mirror pair: equal frequency,
opposite rates (150 of 150)
PASS the whole is balanced at every gain: total rate zero (200 of
200)
E · the actuated zeros: sign changes of Hardy's Z on the line,
heights 10 to 100
sign changes on (10, 100]: 29; theta(100)/pi + 1 = 29.0024
first zero at t = 14.134725142; max error on the first ten:
3.28E-10
max |Im e^{i theta} zeta(1/2+it)| on the scan (Z is real):
4.61E-11
PASS Z(t) = e^{i theta} zeta(1/2 + it) is real on the scan to 1e-8
PASS 29 sign changes of Z on (10, 100]: 29 zeros located on the
line by the executed chart
PASS the count agrees with the smooth term theta(100)/pi + 1,
rounded (29)
PASS the first ten located heights match the tabulated zeros to
1e-8
F · the hardware: generations, forced hypercharges, charges

```

```

write(*,'(a)') 'I · the price of one committed bit at 300 K'
q = KB*300.0_dp*log(2.0_dp)
qexact = 2.8709788850787237e-21_dp
write(*,'(a,es22.15,a)') ' k_B T ln 2 at 300 K = ', q, ' J'
call check('one irreversibly committed bit at 300 K costs
2.870978885e-21 J', abs(q/qexact - 1.0_dp) < 1.0e-14_dp)
end subroutine block_i

pure function bar_t(x) result(y)
integer, intent(in) :: x(3)
integer :: y(3)
y = [-x(1), x(2), x(3)]
end function bar_t

pure function mirror_t(x) result(y)
integer, intent(in) :: x(3)
integer :: y(3)
y = [x(1), x(2), -x(3)]
end function mirror_t

! the Park-Miller minimal standard generator, fixed seed, no
overflow in 64-bit integers
function rnd(s) result(r)
integer(i8), intent(inout) :: s
real(dp) :: r
s = modulo(16807_i8*s, 2147483647_i8)
r = real(s, dp)/2147483647.0_dp
end function rnd

end program substrate_twin

PASS one to six generations: every anomaly vanishes, doublets 4n
(the count is not forced)
PASS at y0 = 1 the five conditions admit one hypercharge vector
in [-12,12]^5: (-4, 2, -3, 6, 3)
PASS charges Q = T3 + Y: up 2/3, down -1/3, neutrino 0, electron
-1
PASS proton one unit, neutron none, hydrogen neutral, vacuum
neutral
PASS the colour lock: of 27 colour triples exactly the 6 on three
distinct axes are nonzero
G · the particle map: every additive charge flipped, the energy kept
PASS the particle map is an involution on the nine-entry table
PASS fixed exactly on the neutral entries: photon, Z, Majorana
neutrino, the dark state
PASS Dirac and Majorana neutrinos share the gauge record and
differ in the one bit
PASS the charge carrier sends the particle map onto the fold (779
of 779)
PASS the two involutions commute; fixed by both exactly when on
the line under both (5453)
PASS the electron and the positron land on the edges h = 0 and h =
2, mirror images
antihydrogen falls with (0.75 +/- 0.13 +/- 0.16) g: from g at 1.21
sigma
PASS gravity reads no charge bit: antimatter falls as matter
within 2 sigma (ALPHA-g)
electron (e -> gamma nu): width < 3.160E-52 eV; width/m <
6.184E-58
proton (p -> e+ pi0): width < 8.691E-58 eV; width/m <
9.262E-67
neutron (free): width = 7.493E-19 eV; width/m =
7.975E-28
muon: width = 2.996E-10 eV; width/m =
2.836E-18
Z boson: width = 2.495E+09 eV; width/m =
2.736E-02
PASS the electron stands on the line to 1e-57 of its rest energy
(partial width bound)
PASS the proton stands on the line to 1e-65 of its rest energy
(partial width bound)
PASS the muon, neutron and Z stand off the line: open modes whose
decay exports to products
H · the neutrino's floor, supplied by deed: oscillation returns
positive mass-squared differences
lightest possible m2 = 8.678E-03 eV; m3 >= 5.030E-02 eV (normal
ordering)
PASS both measured splittings are positive, so at least two
neutrino masses are nonzero
I · the price of one committed bit at 300 K
k_B T ln 2 at 300 K = 2.870978885078724E-21 J
PASS one irreversibly committed bit at 300 K costs
2.870978885e-21 J
BATTERY-JSON: {"checks":34,"failures":0}

```

APPENDIX C: THE SEAT AND THE JUDGMENT

The boot of the session, PhysOS^T 1.0.5p in quick mode, and the engine's screen and judgment.

```
PhysOST 1.0.5p · RECEIPT · mode quick
TOOLCHAIN Lean (version 4.19.0, x86_64-unknown-linux-gnu, commit 6caae842e94, Release) · GNU Fortran (Ubuntu 13.3.0-6ubuntu2-24.04.1) 13.3.0
MANIFEST sha256 a4108cd26bcbe6ef, the files that ran
GROUND toolchain: Lean 4.19.0 and gfortran present · manifest: 31 files signed, all verified · proofs: 2 PhysOS Proofs read from II.10, record
lines complete · source screen: 15 kernels, 23 named premises, clean · citations: 253 laws named in Parts I and II, 253 resolved · quarantine: 6
retirements string-scanned, 3 held by review, clean
EXITS ground 0 · registerA RA_TOE_Thesis 0 · registerB Bridge_Final 0 · registerB Universal_Closure 0 · registerB One_Cut 0 · registerB
One_Cut_Resolution 0 · registerB One_Cut_Terminal 0 · registerB Only_The_Arrow_Remains 0 · registerB Physical_Closure 0 · registerB Forced_Closure
0 · registerB Closure_Executed 0 · registerB Armed_Seat 0 · registerB Census 0 · registerB Heat_Bridge 0 · registerB The_Loop 0 · registerB
RH_Least_Erasure 0 · witness One_Cut_Twin 0 · witness Physical_Closure_Twin 0 · witness Census_Twin 0 · witness Loop_Twin 0 · witness RH_Seal_Twin
0
REGISTER A {"checks":1123,"failures":0,"mode":"sealed"}
WITNESS 5 of 5 twins ran, each held to one battery line with zero failures
CHAIN · D0 83f872725ad9 -> D1 90fc702bbd1 -> D2 492e519daab9 -> D3 6618bec5164b
ELAPSED ground 1 s · Register A 3 s · Register B 34 s · witness 6 s · total 44 s (not hashed)
SEAT EARNED · this run

SCREEN forbidden: none | imports: 0 | axioms declared: 0
laws: 80
JUDGMENT refused 80 of 80 as proof failures | survived none | parse failures none | planted vacuous law survives its negation: True
```